

The Gravitational Two-Body Problem

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Summary

The gravitational two-body problem is defined and described. The classical Keplerian solution for the motion of two point masses is just one specialized version of this problem, and in general the only one which is completely integrable. This chapter will provide a general definition of the two-body problem making no assumptions on the form of the mass distributions that are mutually gravitating. Then various levels of approximation will be introduced, describing the constraints and general results which exist for this problem. The general, or full, two-body problem actually couples rotational and translational motion in the general case, forming a non-integrable problem. Despite this, there are strong constraints on the Hill and impact stability of this problem. In addition, relative equilibria and their stability can be discussed in a general setting. The chapter culminates in a derivation of the Keplerian solution for the dynamics of two point mass bodies orbiting each other.

1 Introduction

The most basic problem in celestial mechanics is the gravitational two-body problem, specifically the motion of two mutually attracting mass distributions. The understanding of the simplest version of this problem occurred with Copernicus' placement of the sun at the center of the solar system and Kepler's subsequent first description of the elliptical motion of the planets in the solar system. What Kepler discovered based on observation and deduction was a special solution to the general gravitational two-body problem, where the bodies in motion can be approximated as spheres or point masses. Kepler's solutions provided the descriptive geometry of planetary motion. These results were not placed onto a firm mechanical basis until Newton's celebrated development of the law of gravitation, his laws of motion, and the calculus. Taken together these advancements provided a complete development of Kepler's solutions based on physical principles. Beyond this significant advance, Newton's work also enabled the complete gravitational two-body problem to be fully posed and analyzed. Details of the dynamics and solutions of the unapproximated problem still required significant development and insight, represented by many of the greats of celestial mechanics, physics and mathematics.

This chapter addresses this most fundamental problem in a general setting, placing an emphasis on recent scholarship and on developing a unified view of the two-body problem. Most striking, it is noted that the full two-body problem is not solvable in most of its statements, and can only be solved under some very restrictive approximations consistent with Kepler's original solution. A complete derivation of Kepler's solution is given at the close of the chapter, but first we develop and focus on the more general statement of this problem and detail the constraints, solutions and approximations that

enable these problems to be understood. This general approach has been motivated by recent scholarship on the general, or so-called full, two-body problem that accounts for the coupled rotational and translational motion of the two mass distributions. Early work that focused on the coupling between translational and rotational motion traces back to Cassini's Laws on the motion of the moon, although in these cases the librational rotation is driven by the orbit, while the influence of the rotation on the orbit is generally neglected. Starting in the 1950s, Duboshin studied the dynamics of coupled rotational and translational motion and stated the general set of differential equations and their fundamental integrals of motion. In the 1970s Kinoshita developed perturbation theories for these problems under the assumption of relatively weak coupling between the different modes of motion. In the 1990s research into the generalized version of this problem was explored by a number of researchers. Wang, Krishnaprasad, and Maddocks approached the problem from a geometrical mechanics approach and explored the existence and stability of relative equilibria, albeit under the assumption that one of the bodies was a sphere. Maciejewski considered the fully general problem and explored the properties of relative equilibria as well as several different ways to pose the problem. In the 2000s a series of articles by Scheeres explored constraints on the solutions to the full two-body problem and applied this problem to the dynamics of binary asteroids and the evolution of rubble pile asteroids. Significant advances in the study of the averaged full two-body problem was made by Boué and Laskar, generalizing the classical Cassini states and showing how they fit into a larger set of integrable motions for the averaged problem.

The approach taken in this article is focused on sharp results that do not make strong assumptions on the motion, such as are found in averaging theories. Solutions which yield general and particular solutions to the problem are given, and constraints for the system which act on the full solution space are emphasized. The goal of the current article is to develop a consistent and unified approach to this problem, relying on classical mechanics formulations. Several theorems are introduced and developed to capture key results that hold for the general system and some special subsystems. There are several results for the general full two-body problem that are not well known and are somewhat surprising and at odds with the classical Kepler solution. First is that the full two-body problem is in general non-integrable and can exhibit chaotic motion (although we will not establish these results here). This occurs either due to the coupling of rotational motion of the two bodies with their relative translational motion or due to the non-spherical mass distributions of either one of the bodies. Second, for the general evolution of the two-sphere problem is that a system with a fixed value of angular momentum can have multiple circular orbits, some of which can be unstable. Only when the system is limited to two point-mass distributions does it become a fully integrable problem. In the following we work through a number of these different results and only arrive at the integrable version of the problem in the last section.

2 Body and Mass Distribution Specifications

The core assumption we make in this chapter is that the mass distributions of the two bodies are rigid, meaning that we do not account for any deformation in their shape or mass distribution. Figure 1 provides a graphical definition of the problem, with the following section providing a mathematical description.

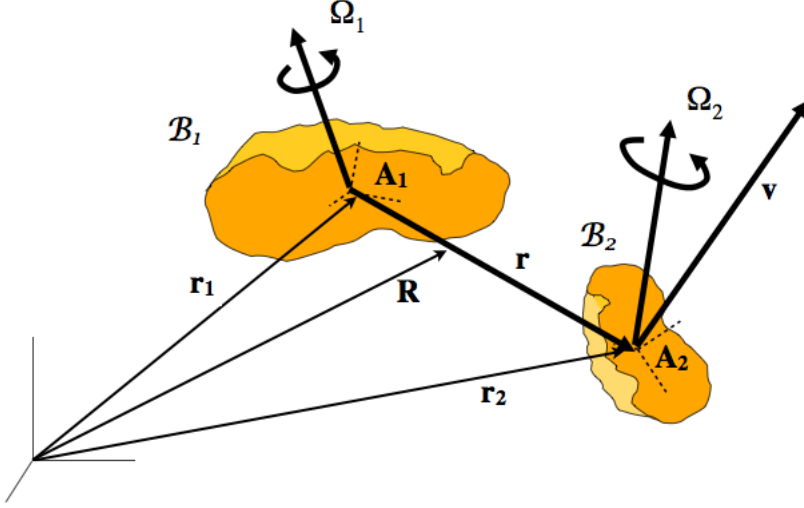


Figure 1: The full gravitational 2-Body Problem and its degrees of freedom.

2.1 Mass and Center of Mass

Assume there exist two rigid bodies with distributed masses, characterized by having finite densities and well defined limits. Their differential masses are defined as

$$dm_i = \sigma_i(\boldsymbol{\rho}) dV \quad (1)$$

where σ_i is the varying density of the i th body, $i = 1, 2$, and $\boldsymbol{\rho}$ is the position of the mass element referenced to some frame. The density is zero when $\boldsymbol{\rho}$ is taken outside of the body and is finite within the body, denoted as $\mathcal{B}_i$. With this definition the total mass of each body equals

$$M_i = \int_{\mathcal{B}_i} dm_i \quad (2)$$

and the location of each of the body's center of mass is computed as

$$\mathbf{r}_i = \frac{1}{M_i} \int_{\mathcal{B}_i} \boldsymbol{\rho} dm_i \quad (3)$$

The joint mass distribution of the system is defined via the differential mass element

$$dm = dm_1 + dm_2 \quad (4)$$

and the joint bodies as $\mathcal{B} = \{\mathcal{B}_1, \mathcal{B}_2\}$. This allows the total mass of the system to be defined as

$$M_1 + M_2 = \int_{\mathcal{B}} dm \quad (5)$$

The barycenter of the system is then found as

$$\mathbf{R} = \frac{1}{M_1 + M_2} \int_{\mathcal{B}} \boldsymbol{\rho} dm \quad (6)$$

As will be discussed later, we can take $\mathbf{R} \equiv \mathbf{0}$ in general.

2.2 Relative Orientations

The orientation of the bodies is defined by rotation dyadics that transform a vector in the body-fixed frame into inertial space, denoted as $\mathbf{A}_1$ and $\mathbf{A}_2$. Then the complete specification of each body in an inertial frame is $\{\mathbf{r}_1, \mathbf{r}_2, \mathbf{A}_1, \mathbf{A}_2\}$ and the relative position and orientation of the bodies with respect to each other is $\{\mathbf{r}, \mathbf{A}\}$ where the relative position and attitude of body 2 relative to body 1 is defined as

$$\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1 \quad (7)$$

$$\mathbf{A} = \mathbf{A}_1^T \cdot \mathbf{A}_2 \quad (8)$$

As each of these terms represent 3 degrees of freedom, to specify the relative position and orientation of two rigid bodies requires a total of 6 degrees of freedom. These are called the internal or relative degrees of freedom. To orient these internal degrees of freedom with respect to an inertial frame requires an additional 3 degrees of freedom, represented by the rotation dyadic $\mathbf{A}_1$. Figure 1 shows a graphical representation of this full system. We note that each of the rigid bodies has an angular velocity $\boldsymbol{\Omega}_i$ that defines the instantaneous rate of rotation between the body-fixed frame and an inertial frame.

2.3 Moments of Inertia

An additional mass distribution quantity of interest are the inertia dyadics of each body, defined by

$$\mathbf{I}_i = \int_{\mathcal{B}_i} [\rho^2 \mathbf{U} - \boldsymbol{\rho} \boldsymbol{\rho}] dm_i \quad (9)$$

where $\mathbf{U}$ is the identity dyadic and the product of two vectors, e.g. $\boldsymbol{\rho} \boldsymbol{\rho}$, is a dyad. The coefficients of this dyadic can be computed in closed form for special cases such as constant density spheres, ellipsoids, and polyhedra.

The inertia dyadic of the entire system can also be specified as

$$\mathbf{I} = \int_{\mathcal{B}} [\rho^2 \mathbf{U} - \boldsymbol{\rho} \boldsymbol{\rho}] dm \quad (10)$$

and simplifies into the individual inertia dyadics and the inertia dyadic of the two masses if the system is computed relative to the barycenter

$$\mathbf{I} = \frac{M_1 M_2}{M_1 + M_2} r^2 [\mathbf{U} - \hat{\mathbf{r}} \hat{\mathbf{r}}] + \mathbf{I}_1 + \mathbf{I}_2 \quad (11)$$

Of interest later is the moment of inertia relative to a fixed direction, $\hat{\mathbf{H}}$, computed as

$$I_H = \hat{\mathbf{H}} \cdot \mathbf{I} \cdot \hat{\mathbf{H}} \quad (12)$$

Note that I_H is a function of the relative position of the two masses and their orientation, all relative to the unit vector $\hat{\mathbf{H}}$.

Also of interest is the polar moment of inertia, computed as one-half of the trace of the total inertia

$$I_P = \frac{1}{2} \text{Trace}(\mathbf{I}) \quad (13)$$

$$= \frac{M_1 M_2}{M_1 + M_2} r^2 + \frac{1}{2} \text{Trace}(\mathbf{I}_1 + \mathbf{I}_2) \quad (14)$$

Note that the polar moment of inertia for a 2-body system is only a function of the separation between the bodies, or $I_P(r)$, and is independent of the orientation of the two mass distributions. An important inequality can be proven between the polar moment of inertia and the moment of inertia relative to a fixed direction

$$I_H \leq I_P \quad (15)$$

which holds for any fixed direction $\hat{\mathbf{H}}$.

2.4 Body Shapes and Geometry

In the following we will assume that both bodies are convex. This is not an essential assumption but makes it simpler to discuss situations when the two bodies can come into contact. If they are both convex, then at every relative configuration of the system there is a well defined minimum distance between the bodies $d(\hat{\mathbf{r}}, \mathbf{A})$ defined as the radius at which the two bodies touch for their relative configuration. This distance changes smoothly with $\hat{\mathbf{r}}$ and $\mathbf{A}$ and is constant for all rotations of $\mathbf{A}$ about the unit vector $\hat{\mathbf{r}}$. If non-convex bodies are assumed then there is the potential for multiple distances between the distributions at a given relative configuration and discontinuities in the minimum distance as a function of $\hat{\mathbf{r}}$ and $\mathbf{A}$.

The maximum of these minimum distances can be specified as $D = \max_{\hat{\mathbf{r}}, \mathbf{A}} d(\hat{\mathbf{r}}, \mathbf{A})$. This quantity is of fundamental interest for any two body system as beyond this distance the two bodies can never impact with each other. This limit can be defined independent of whether the bodies are convex or not.

3 Newtonian Gravitational Attraction

Having defined the two bodies and their mass distribution, we next consider the relative forces that these distributions apply to each other due to Newton's law of gravitational attraction.

3.1 Relative Forces

Newton's fundamental law of gravitational attraction states that two differential mass elements will experience an attraction between them proportional to the product of the masses, inversely proportional to the square of the distance between them and directed along their relative position. Given our definition of the relative position vector $\mathbf{r}$ as going from body 1 to body 2 the differential force that a mass element in body 1 places on a mass element in body 2 is

$$d\mathbf{F}_{12} = -\frac{\mathcal{G}dm_1 dm_2}{|\mathbf{r} + \mathbf{A}_2 \cdot \boldsymbol{\rho}_2 - \mathbf{A}_1 \cdot \boldsymbol{\rho}_1|^3} (\mathbf{r} + \mathbf{A}_2 \cdot \boldsymbol{\rho}_2 - \mathbf{A}_1 \cdot \boldsymbol{\rho}_1) \quad (16)$$

where we recall that $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$ and $\boldsymbol{\rho}_i$ is the position of a mass element of body i relative to its center of mass. The differential force of a mass element in body 2 on a mass element in body 1 is simply $-d\mathbf{F}_{12}$ in accordance with Newton's law of action and reaction. To compute the total force that body 1 exerts on body 2, and vice-versa, these

differentials are integrated over both mass distributions:

$$\mathbf{F}_{12} = \int_{\mathcal{B}_1} \int_{\mathcal{B}_2} d\mathbf{F}_{12} \quad (17)$$

and $\mathbf{F}_{21} = -\mathbf{F}_{12}$.

The differential force can also be derived from a scalar potential function, defined as the differential potential gravitational energy between the bodies

$$dU = - \frac{\mathcal{G} dm_1 dm_2}{|\mathbf{r} + \mathbf{A}_2 \cdot \boldsymbol{\rho}_2 - \mathbf{A}_1 \cdot \boldsymbol{\rho}_1|} \quad (18)$$

then

$$d\mathbf{F}_{12} = - \frac{\partial dU}{\partial(\mathbf{r}_2 + \boldsymbol{\rho}_2)} \quad (19)$$

$$d\mathbf{F}_{21} = - \frac{\partial dU}{\partial(\mathbf{r}_1 + \boldsymbol{\rho}_1)} \quad (20)$$

Note that $\frac{\partial dU}{\partial(\mathbf{r}_i + \boldsymbol{\rho}_i)} = \frac{\partial dU}{\partial \mathbf{r}_i}$ in general.

The gravitational potential energy between the bodies is then defined by integrating this differential potential over both bodies

$$U(\mathbf{r}_1, \mathbf{r}_2, \mathbf{A}_1, \mathbf{A}_2) = \int_{\mathcal{B}_1} \int_{\mathcal{B}_2} dU \quad (21)$$

$$- \mathcal{G} \int_{\mathcal{B}_1} \int_{\mathcal{B}_2} \frac{dm_1(\boldsymbol{\rho}_1) dm_2(\boldsymbol{\rho}_2)}{|\mathbf{r}_2 - \mathbf{r}_1 + \mathbf{A}_2 \cdot \boldsymbol{\rho}_2 - \mathbf{A}_1 \cdot \boldsymbol{\rho}_1|} \quad (22)$$

where the integration variables $\boldsymbol{\rho}_i$ are expressed in their respective body-fixed frames and all terms in the vector magnitude are specified in inertial space. The forces of these bodies on each other are then computed as

$$\mathbf{F}_{12} = - \frac{\partial U}{\partial \mathbf{r}_2} \quad (23)$$

$$\mathbf{F}_{21} = - \frac{\partial U}{\partial \mathbf{r}_1} \quad (24)$$

3.2 Relative Moments

For bodies with finite mass distributions it is also necessary to compute the mutual moments of the force (or moments) that are exerted on each other. The moment that

body 1 exerts on body 2 is found by integrating the differential moment over both bodies:

$$d\mathbf{M}_{12} = \mathbf{A}_2 \cdot \boldsymbol{\rho}_2 \times d\mathbf{F}_{12} \quad (25)$$

$$\mathbf{M}_{12} = \int_{\mathcal{B}_1} \int_{\mathcal{B}_2} \mathbf{A}_2 \cdot \boldsymbol{\rho}_2 \times d\mathbf{F}_{12} \quad (26)$$

The moment can also be related to the mutual potential, although the details are more involved. This is found by taking the partial of the mutual potential with respect to the infinitesimal rotations about each axis of body 2, which we represent as $\mathbf{M}_{12} = -U_{\boldsymbol{\theta}_2}$. In practice, once the relative attitude of body 2 is defined explicitly using $\mathbf{A}_2$ the partials with respect to the angles can be computed from this relationship. The moment acting on body 1, $\mathbf{M}_{21}$, is similarly computed as $-U_{\boldsymbol{\theta}_1}$. It must be noted that the mutual moments are not equal and opposite, but that their sum equals the negative moment of the total gravitational force between the bodies

$$\mathbf{M}_{12} + \mathbf{M}_{21} + (\mathbf{r}_2 - \mathbf{r}_1) \times \mathbf{F}_{12} = 0 \quad (27)$$

meaning that we do not need to solve for one of the moments.

3.3 Internal Configuration

It is also possible to express the potential completely in terms of the internal configuration of the system relative to body-fixed frame of $\mathcal{B}_1$, for example, $\mathbf{r} = \mathbf{A}_1^T \cdot (\mathbf{r}_2 - \mathbf{r}_1)$ and $\mathbf{A} = \mathbf{A}_1^T \cdot \mathbf{A}_2$.

$$U(\mathbf{r}, \mathbf{A}) = -\mathcal{G} \int_{\mathcal{B}_1} \int_{\mathcal{B}_2} \frac{dm_1(\boldsymbol{\rho}_1) dm_2(\boldsymbol{\rho}_2)}{|\mathbf{r} + \mathbf{A} \cdot \boldsymbol{\rho}_2 - \boldsymbol{\rho}_1|} \quad (28)$$

Then the forces of these bodies on each other are then computed as

$$\mathbf{F}_{12} = -\frac{\partial U}{\partial \mathbf{r}} \quad (29)$$

$$\mathbf{F}_{21} = \frac{\partial U}{\partial \mathbf{r}} \quad (30)$$

and the moments are computed as

$$\mathbf{M}_{12} = -\frac{\partial U}{\partial \boldsymbol{\theta}} \quad (31)$$

$$\mathbf{M}_{21} = \frac{\partial U}{\partial \boldsymbol{\theta}} + \mathbf{r} \times \frac{\partial U}{\partial \mathbf{r}} \quad (32)$$

where $\boldsymbol{\theta}$ is implicitly tied to the attitude variations found in $\mathbf{A}$.

3.4 Bounds on the Gravitational Potential

Following are a number of useful bounds on the gravitational potential that will be appealed to in our later discussions. These results are all proven and discussed in more detail in the literature. Before we start it is trivial to note that the gravitational potential energy of the system is never positive, or that $U \leq 0$ for all internal configurations.

A useful concept is the minimum potential at a given, fixed distance between the bodies

$$U_m(r) = \min_{\mathbf{r}, \mathbf{A}} U(\mathbf{r}, \mathbf{A})$$

Thus $U(\mathbf{r}, \mathbf{A}) \geq U_m(r)$ for all separation distances r .

An explicit bound on $U_m(r)$ can be found:

$$U_m(r) \geq \frac{-\mathcal{G}M_1M_2}{r-D} \quad \text{if } r > D \quad (33)$$

where D is the maximum contact distance between the bodies and was defined in Section 2.4. Thus, upper and lower bounds on the potential energy can be found, $\frac{-\mathcal{G}M_1M_2}{r-D} \leq U \leq 0$. From these it can be immediately noted that $\lim_{r \rightarrow \infty} U \rightarrow 0$.

It can also be shown that the gravitational potential is monotonic in the distance between the bodies. This can be established in two different ways. First, given two parallel vectors, $\mathbf{r} = r\hat{\mathbf{r}}$ and $\mathbf{d} = d\hat{\mathbf{r}}$ where $d < r$, if $r + d > 2D$ then it can be proven that $U(\mathbf{r}, \mathbf{A}) \geq U(\mathbf{d}, \mathbf{A})$. A second monotonic result can be developed as the distance between the bodies grows larger. Specifically, a generalization of Euler's Theorem on homogenous functions can be developed for the gravitational potential that establishes that the gravitational potential is strictly increasing with the distance between the bodies

$$0 < \frac{\partial U}{\partial \mathbf{r}} \cdot \mathbf{r} < -2U \quad (34)$$

for $r > 2D$.

3.5 Specialized Mass Distributions

Analytical statements of the general mutual potential are difficult. Thus, there are several different approximate statements of this potential function that have been developed. These are all generally tied to the geometric shape of the mass distribution in conjunction with a constant density assumption. A few of the more important ones are detailed in the following.

Finite Density Sphere and a General Shape An important simplification occurs if one of the bodies is a sphere (or point mass) while the other (body 1, say) remains a general mass distribution. Then, the mutual potential can be arbitrarily integrated over the spherical body and reduces to

$$U(\mathbf{r}) = -\mathcal{G}M_2 \int_{\mathcal{B}_1} \frac{dm_1(\boldsymbol{\rho}_1)}{|\mathbf{r} - \boldsymbol{\rho}_1|} \quad (35)$$

which is independent of the attitude of the sphere (body 2). Geometrically this occurs as the attitude between a general body and a point can be reduced to the relative position between the two of them. Thus, $\mathbf{M}_{12} = 0$ and $\mathbf{M}_{21} = \mathbf{r} \times \frac{\partial U}{\partial \mathbf{r}}$. This formulation can also take advantage of the well developed theory for the gravitational potential of a single body, where there are spherical harmonic expansion forms for the general case of a mass distribution, and also analytical forms for special shapes such as ellipsoids and polyhedra.

Finite Density Spheres If the two mass distributions are assumed to be rigid spheres of a given density, then it is a classical result to show that the mutual potential reduces to that of the two point mass potential, even though now the bodies have non-zero moments of inertia and, especially when in contact, can be forced to spin. Accordingly, the bodies have moments of inertia which are equal across any axis. For a sphere of diameter d they are simply

$$I = \frac{1}{10}Md^2 \quad (36)$$

For two spheres of diameters d_1 and d_2 the minimum distance between them is just the average of the diameters: $D = (d_1 + d_2)/2$.

Point Mass Bodies In the case where the two density distributions correspond to points, the potential and equations simplify. In this case $dm_i = M_i\delta(|\mathbf{r} - \mathbf{r}_i|)$, where $\delta(-)$ is the Kronecker delta function. Then the mutual potential devolves to

$$U = -\frac{\mathcal{G}M_1M_2}{r} \quad (37)$$

with the relative attitude degrees of freedom disappearing. Due to this the two moments M_{12} and M_{21} are zero and, due to the collinearity of the mutual force with the relative position, the relative moment between the bodies also vanishes. Finally, the mutual distance between the bodies can be decreased to $D = 0$ and the moments of inertia of the points are identically zero.

4 Equations of Motion

The full equations of motion for this system can be derived using principles of virtual work. Define the action of the system as the time integral of the Lagrangian, which is the kinetic energy minus the potential energy. The potential energy is given in Eqn. 22 while the kinetic energy can be written as

$$T = \frac{1}{2} \frac{M_1 M_2}{M_1 + M_2} \dot{\mathbf{r}} \cdot \dot{\mathbf{r}} + \frac{1}{2} \boldsymbol{\Omega}_1 \cdot \mathbf{I}_1 \cdot \boldsymbol{\Omega}_1 + \frac{1}{2} \boldsymbol{\Omega}_2 \cdot \mathbf{I}_2 \cdot \boldsymbol{\Omega}_2 \quad (38)$$

where we explicitly make the replacement $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$. The Lagrangian is $L = T - U$ and the action is $\mathcal{A} = \int_{t_0}^{t_1} L \, d\tau$. The equations of motion can be derived from d'Alembert's principle by finding the conditions to form a stationary value of the action between two fixed end orientations. Consider variations in each of the degrees of freedom, $\mathbf{r}$, $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$, where the angular variations are just infinitesimal variations in the attitudes of each body about a specified $\mathbf{A}_1$ or $\mathbf{A}_2$. This leads to

$$\delta \mathcal{A} = \int_{t_0}^{t_1} (\delta T - \delta U) \, d\tau \quad (39)$$

$$= \int_{t_0}^{t_1} \left[\frac{M_1 M_2}{M_1 + M_2} \dot{\mathbf{r}} \cdot \delta \dot{\mathbf{r}} - U_{\mathbf{r}} \cdot \delta \mathbf{r} + \sum_{i=1}^2 \boldsymbol{\Omega}_i \cdot \mathbf{I}_i \cdot \delta \boldsymbol{\Omega}_i - U_{\boldsymbol{\theta}_i} \cdot \delta \boldsymbol{\theta}_i \right] d\tau \quad (40)$$

Note that $\delta(d\mathbf{r}/dt) = d(\delta\mathbf{r})/dt$ and that $\delta\boldsymbol{\Omega} = \delta d\boldsymbol{\theta}/dt = d(\delta\boldsymbol{\theta})/dt$. Then integration by parts can be applied to find

$$\begin{aligned} & \int_{t_0}^{t_1} \left[\frac{M_1 M_2}{M_1 + M_2} \dot{\mathbf{r}} \cdot d(\delta\mathbf{r})/d\tau + \sum_{i=1}^2 \boldsymbol{\Omega}_i \cdot \mathbf{I}_i \cdot d(\delta\boldsymbol{\theta}_i)/d\tau \right] d\tau \\ &= \left[\frac{M_1 M_2}{M_1 + M_2} \dot{\mathbf{r}} \cdot \delta \mathbf{r} + \sum_{i=1}^2 \boldsymbol{\Omega}_i \cdot \mathbf{I}_i \cdot \delta \boldsymbol{\theta}_i \right]_{t_0}^{t_1} - \int_{t_0}^{t_1} \left[d\left(\frac{M_1 M_2}{M_1 + M_2} \dot{\mathbf{r}}\right)/d\tau \cdot \delta \mathbf{r} + \sum_{i=1}^2 d(\boldsymbol{\Omega}_i \cdot \mathbf{I}_i)/d\tau \cdot \delta \boldsymbol{\theta}_i \right] d\tau \end{aligned} \quad (41)$$

Setting the end point variations to zero the full variation of the action is found to be

$$\delta \mathcal{A} = - \int_{t_0}^{t_1} \left[\left(d\left(\frac{M_1 M_2}{M_1 + M_2} \dot{\mathbf{r}}\right)/d\tau + U_{\mathbf{r}} \right) \cdot \delta \mathbf{r} + \sum_{i=1}^2 \left(d(\boldsymbol{\Omega}_i \cdot \mathbf{I}_i)/d\tau + U_{\boldsymbol{\theta}_i} \right) \cdot \delta \boldsymbol{\theta}_i \right] d\tau \quad (42)$$

From the principle of least action the unique equations of motion can be extracted for this system

$$\frac{M_1 M_2}{M_1 + M_2} \ddot{\mathbf{r}} = -U_{\mathbf{r}} \quad (43)$$

$$\mathbf{A}_i \left[\mathbf{I}_i \cdot \dot{\boldsymbol{\Omega}}_i + \boldsymbol{\Omega}_i \times \mathbf{I}_i \cdot \boldsymbol{\Omega}_i \right] = -U_{\boldsymbol{\theta}_i} ; \, i = 1, 2 \quad (44)$$

These equations are all stated in an inertial frame, although the time derivatives for the rotational equations are chosen in their respective body frames. A statement for the kinematics of rotation must also be given. With our assumption that the attitude dyadics $\mathbf{A}_i$ map from a body-fixed to an inertial frame, the kinematical equation for them is simply stated as

$$\dot{\mathbf{A}}_i = \mathbf{A}_i \cdot \tilde{\boldsymbol{\Omega}}_i \quad (45)$$

where the $\tilde{}$ operator defines the cross product dyadic. Specifically, if $\mathbf{a} = a_1\hat{\mathbf{e}}_1 + a_2\hat{\mathbf{e}}_2 + a_3\hat{\mathbf{e}}_3$ then $\tilde{\mathbf{a}} = a_1(\hat{\mathbf{e}}_3\hat{\mathbf{e}}_2 - \hat{\mathbf{e}}_2\hat{\mathbf{e}}_3) + a_2(\hat{\mathbf{e}}_1\hat{\mathbf{e}}_3 - \hat{\mathbf{e}}_3\hat{\mathbf{e}}_1) + a_3(\hat{\mathbf{e}}_2\hat{\mathbf{e}}_1 - \hat{\mathbf{e}}_1\hat{\mathbf{e}}_2)$.

The above equations represent the most general case of the rigid two body problem. In addition to this most general form of the equations, there are a few specific simplifications of interest motivated by different physical models. These are considered in the following.

4.1 Restricted Problem

If one of the bodies has zero total mass, then it no longer affects the other body, but is still affected by the other body. The body with mass then rotates in a moment free environment which can be solved in closed form. The motion of the massless particle is still majorly influenced by the tumbling or rotating body.

4.2 Sphere Restricted Assumption

If we assume that one of the bodies is a finite sphere we note that the rotational dynamics of that body are trivial and drop out of the equation. Assume that body $i = 2$ is the sphere, then $\mathbf{I}_2 = I_2\mathbf{U}$ where $\mathbf{U}$ is the unity dyadic and I_2 is the moment of inertia of the sphere. Then we note that $\boldsymbol{\Omega}_2 \times \mathbf{I}_2 \cdot \boldsymbol{\Omega}_2 = 0$, and from symmetry, that $U_{\boldsymbol{\theta}_2} = 0$. Thus the rotational equations for body two are just $I_2\dot{\boldsymbol{\Omega}}_2 = \mathbf{0}$ and it rotates at its initial rate and direction without change. Furthermore, the gravitational potential can be explicitly integrated over the spherical body, yielding a simplified form of the mutual potential as given in Eqn. 35.

4.3 Finite Density Sphere Dynamics

If we assume that both bodies are finite density spheres, both of the rotational equations decouple and both bodies spin at a constant rate. The mutual potential is now independent of the body attitudes and the translational dynamics reduce to the classical Kepler

2-body problem. The fact that both bodies have finite densities means that there is a minimum distance between them and that both bodies still have non-zero moments of inertia. If the spheres touch, assuming friction, or distort slightly under mutual gravitational attraction, due to tides, it will be seen that this system has a much different set of minimum energy states than the point mass problem.

4.4 Point Mass Translational Dynamics

Finally, if both bodies are modeled as point masses their moments of inertia and lower separation distance bound both go to zero. As before, the equations for translational motion can be integrated in closed form and represent the Keplerian solution (see Section 8).

5 Conservation Principles and Constraints

There are a number of constraints and conservation principles that control the solutions to the general equations of motion.

5.1 General Integrals of Motion

The general gravitational 2-Body problem has three important conservation principles: conservation of linear momentum, conservation of angular momentum and conservation of energy. Conservation of momentum only relies on the absence of exogenous forces and moments, and thus is more fundamental. Conservation of energy relies on both the absence of exogenous forces and moments and on the absence of internal, non-potential (dissipative) forces and moments due to interactions such as friction. As almost all real celestial mechanics problems have some level of internal dissipation, we have a particular interest in understanding the minimum energy states of the two body problem. This is discussed in a following chapter.

5.1.1 Linear Momentum

Conservation of linear momentum for the general 2-Body system is stated in terms of the system momentum:

$$\mathbf{P} = M_1 \dot{\mathbf{r}}_1 + M_2 \dot{\mathbf{r}}_2 \quad (46)$$

where $\dot{\mathbf{r}}_i$ is the velocity of the center of mass of body i . The time derivative of this expression yields

$$\dot{\mathbf{P}} = \mathbf{F}_{21} + \mathbf{F}_{12} \quad (47)$$

but these forces are equal and opposite, yielding $\dot{\mathbf{P}} = 0$. Defining the center of mass of the system as

$$\mathbf{R} = \frac{1}{M_1 + M_2} (M_1 \mathbf{r}_1 + M_2 \mathbf{r}_2) \quad (48)$$

we can then explicitly solve for the motion of the system center of mass

$$\mathbf{R}(t) = \mathbf{R}_o + \dot{\mathbf{R}}_o(t - t_o) \quad (49)$$

Transforming our coordinate system to this uniformly moving center of mass is equivalent to our initial decision to take the system barycenter as the origin.

5.1.2 Angular Momentum

The total angular momentum of the system can be expressed in terms of the relative position and velocity and the rotational angular momentum of each body as

$$\mathbf{H} = \frac{M_1 M_2}{M_1 + M_2} \mathbf{r} \times \dot{\mathbf{r}} + \mathbf{A}_1 \cdot \mathbf{I}_1 \cdot \boldsymbol{\Omega}_1 + \mathbf{A}_2 \cdot \mathbf{I}_2 \cdot \boldsymbol{\Omega}_2 \quad (50)$$

Taking the time derivative of this quantity yields

$$\dot{\mathbf{H}} = \frac{M_1 M_2}{M_1 + M_2} \mathbf{r} \times \ddot{\mathbf{r}} + \sum_{i=1}^2 \mathbf{A}_i \cdot \left[\tilde{\boldsymbol{\Omega}}_i \cdot \mathbf{I}_i \cdot \boldsymbol{\Omega}_i + \mathbf{I}_i \cdot \dot{\boldsymbol{\Omega}}_i \right] \quad (51)$$

$$= -\mathbf{r} \times U_{\mathbf{r}} - \sum_{i=1}^2 U_{\boldsymbol{\theta}_i} \quad (52)$$

$$\equiv 0 \quad (53)$$

the last step arising from Eqn. 27. We note that if there are any internal forces or torques acting on the system, that these will always occur in equal and opposite pairs and will naturally cancel out.

5.1.3 Energy

Finally, consider the total energy of the system

$$E = T + U \quad (54)$$

Direct differentiation of this expression yields

$$\dot{E} = -\dot{\mathbf{r}} \cdot \mathbf{U}_{\mathbf{r}} - \sum_{i=1}^2 \boldsymbol{\Omega}_i \cdot \mathbf{U}_{\boldsymbol{\theta}_i} + \mathbf{U}_{\mathbf{r}} \cdot \dot{\mathbf{r}} + \sum_{i=1}^2 \mathbf{U}_{\boldsymbol{\theta}_i} \cdot \boldsymbol{\Omega}_i \quad (55)$$

$$\equiv 0 \quad (56)$$

Now we note that if there are any additional internal forces or torques due to dissipative effects, that these may not automatically cancel due to their dot product with the translational or rotational velocities. Due to this, the total energy of a 2-body system may not always be conserved but can decrease in many instances.

5.2 Modified Sundman's Inequality

In addition to these integrals of motion, it is possible to derive an important constraint on these systems. To start, note that $\mathbf{H} = H \hat{\mathbf{H}}$, where H is the magnitude and $\hat{\mathbf{H}}$ is the unit vector direction that the constant angular momentum points, comprising two of the three integrals associated with the angular momentum. Then the following inequality can be developed

$$H = \hat{\mathbf{H}} \cdot \mathbf{H} \quad (57)$$

$$= \int_{\mathcal{B}} \hat{\mathbf{H}} \cdot (\boldsymbol{\rho} \times \dot{\boldsymbol{\rho}}) dm(\boldsymbol{\rho}) \quad (58)$$

$$= \int_{\mathcal{B}} \dot{\boldsymbol{\rho}} \cdot (\hat{\mathbf{H}} \times \boldsymbol{\rho}) dm(\boldsymbol{\rho}) \quad (59)$$

$$\leq \int_{\mathcal{B}} |\dot{\boldsymbol{\rho}}| |\hat{\mathbf{H}} \times \boldsymbol{\rho}| dm(\boldsymbol{\rho}) \quad (60)$$

The last step arises from the triangle inequality. Now recall the Cauchy-Schwarz inequality $[\int AB dx]^2 \leq [\int A^2 dx] [\int B^2 dx]$. This leads to

$$H^2 \leq \left[\int_{\mathcal{B}} |\dot{\boldsymbol{\rho}}| |\hat{\mathbf{H}} \times \boldsymbol{\rho}| dm(\boldsymbol{\rho}) \right]^2 \quad (61)$$

$$\leq \left[\int_{\mathcal{B}} \dot{\boldsymbol{\rho}} \cdot \dot{\boldsymbol{\rho}} dm(\boldsymbol{\rho}) \right] \left[\hat{\mathbf{H}} \cdot \int_{\mathcal{B}} \tilde{\boldsymbol{\rho}} \cdot \tilde{\boldsymbol{\rho}} dm(\boldsymbol{\rho}) \cdot \hat{\mathbf{H}} \right] \quad (62)$$

In the final statement the unit vector can be drawn outside of the integral as it is a constant. Then the first integral equals $2T$, twice the kinetic energy, and the second integral is just the system moment of inertia about the angular momentum axis, denoted here as $I_H = \hat{\mathbf{H}} \cdot \mathbf{I} \cdot \hat{\mathbf{H}}$. Thus we find a generalization of the Sundman Inequality of the N -body problem

$$H^2 \leq 2T I_H \quad (63)$$

The classical statement of the Sundman inequality involves the polar moment of inertia, I_P , and is stated as $H^2 \leq 2TI_P$. For distributed body problems, our alternate statement is important and is sharper than the classical result, as $I_H \leq I_P$. The classical statement of the Sundman Inequality is used later to establish stability against impact.

5.3 Minimum Energy Function

The Modified Sundman Inequality can be used to define a lower bound on the system energy as a function of total angular momentum. The derivation is simple, and involves the substitution of $E - U$ for the kinetic energy

$$H^2 \leq 2(E - U)I_H \quad (64)$$

This can be solved for energy and provides a strict limit on the system energy as a function of angular momentum:

$$\frac{H^2}{2I_H} + U \leq E \quad (65)$$

The functional quantity on the left is only a function of the system internal configuration and the body orientation relative to the inertial direction of the angular momentum magnitude. This is essentially Smale's amended potential, but we will call it the minimum energy function in accordance with our use of it. The minimum energy function is defined as

$$\mathcal{E}(\mathbf{r}, \mathbf{A}, \hat{\mathbf{H}}) = \frac{H^2}{2I_H(\mathbf{r}, \mathbf{A}, \hat{\mathbf{H}})} + U(\mathbf{r}, \mathbf{A}) \quad (66)$$

This provides a minimum bound on the energy for every possible system configuration: $\mathcal{E}(\mathbf{r}, \mathbf{A}, \hat{\mathbf{H}}) \leq E$. This simple function enables a complete analysis of all relative equilibrium in the Full 2-Body problem and their related stability, and is discussed in more detail in Section 7.

6 Constraints on Motion: Escape and Impact

There are two major constraints that can be placed on the relative motion of two bodies: whether they undergo mutual escape or mutual impact. For both cases there are strong sufficient conditions that can be placed on the system energy and angular momentum to ensure that either escape, impact or both cannot occur. These questions are dealt with in more detail in the literature.

6.1 Hill Stability

Definition 1 *A two-body system is Hill stable if the distance between the components are bounded for all time.*

For the full two-body problem a sufficient condition for Hill stability is easily established.

Theorem 1 *If the total energy for a two-body system is strictly negative, $E < 0$, then the system is Hill stable. Conversely, if a system undergoes mutual escape then $E \geq 0$.*

Proof 1 *The total energy can be decomposed into a kinetic and potential energy, $E = T + U$. Note the following facts discussed before: $T \geq 0$ and $U \leq 0$ over the interval $r \in (0, \infty)$, $\lim_{r \rightarrow \infty} U(r) = 0$, and $(\partial U / \partial \mathbf{r}) \cdot \mathbf{r} > 0$ for $r > 2D$. For a given separation between two bodies $r > 2D$, it is possible to construct a bounding potential such that $U(r) \geq -\frac{\mathcal{G}M_1M_2}{r-D}$. Then, if the total energy E is negative, the equation $E = -\frac{\mathcal{G}M_1M_2}{R-D}$ can be solved for a unique value of R such that $D < R < \infty$. This leads to the equality $E = T + U = -\frac{\mathcal{G}M_1M_2}{R-D}$. Then $T = -\frac{\mathcal{G}M_1M_2}{R-D} - U(r) \geq 0$, and $-\frac{\mathcal{G}M_1M_2}{R-D} \geq U$. Due to the strong monotonicity of U for $r > 2D$, this yields the constraint $r \leq R < \infty$.*

Conversely, if escape occurs then $E = \lim_{r \rightarrow \infty} (T + U) = T_\infty \geq 0$.

Stronger statements on the conditions for when a system with positive energy will escape are more difficult to develop. In a number of special cases they can be established, such as in the two point mass problem and for somewhat specialized conditions. It is interesting to note that a systematic study of this phenomenon has found that for even small deviations from a spherical shapes, systems started from an unstable relative equilibrium with a positive energy will apparently eventually disrupt.

6.2 Impact Stability

Definition 2 *A two body system is Stable Against Impact (SAI) if the distance between the two mass centers is bounded away from their maximum contact distance for a given time and all future times, or, $r(t) > D$ for $t \in [t_o, \infty)$.*

In classical celestial mechanics Lagrange stability is posed as boundedness of the entire state of the system, including position as well as velocity. This is in contrast to

Hill stability which only refers to bounds on the position. The main difference between these in the classical two point mass problem is that Hill stability allows impact between infinitesimal masses while Lagrange stability does not. For the current case, we note that for a finite density system with a negative energy that the state of the bodies are always bounded and hence, are Lagrange stable. Thus we do not use this terminology here and replace it with the more descriptive Stability Against Impact (SAI).

SAI can be developed in a constructive sense in terms of zero-velocity curves that delineate regions where motion is restricted by the exhaustion of all excess energy, defined as the difference between the current system energy and the minimum energy attainable at the given value of angular momentum. It can be shown that a similar phenomenon controls Hill stability. From a mechanical aspect, SAI occurs whenever there is enough angular momentum in the system to create an inner relative equilibrium point (discussed in the next section), combined with a low enough overall energy value so that a trajectory can be trapped on the outer side of this point. Thus this condition is more involved and involves knowledge of both the angular momentum and energy of the system.

Such a constructive proof can be bypassed, and in the following we present a more general condition for SAI.

Theorem 2 *A binary system with a separation distance at time t_o that satisfies $r_o = r(t_o) > D$, a total angular momentum H , and a total energy E is SAI if a number $R \in (D, r_o)$ can be found such that:*

$$I_P(R) \leq \frac{H^2}{2(E - U_m(R))} \quad (67)$$

where $I_P(R)$ is defined in Section 2.3 and $U_m(R)$ is defined in Section 3.4.

Proof 2 *To prove, rewrite the classical Sundman Inequality $H^2 \leq 2(E - U)I_P$ as:*

$$\frac{H^2}{2(E - U)} \leq I_P \quad (68)$$

If two vectors $\mathbf{r}$ and $\mathbf{R}$ are parallel, then $U(\mathbf{r}, \mathbf{A}) > U(\mathbf{R}, \mathbf{A})$ if $r > R$ and $r + R > 2D$. Then, by definition, $U(\mathbf{r}, \mathbf{A}) > U(\mathbf{R}, \mathbf{A}) \geq U_m(R)$ and for $r > R > D$ we have:

$$\frac{H^2}{2(E - U_m(R))} < \frac{H^2}{2(E - U(\mathbf{r}, \mathbf{A}))} \quad (69)$$

where the term on the left is constant for a given R . From the classical Sundman Inequality we have

$$\frac{H^2}{2(E - U(\mathbf{r}, \mathbf{A}))} \leq I_P(r) \quad (70)$$

which holds for all $r = r(t)$, $t \in [t_o, \infty)$. Then, if the proper value of R can be found to satisfy Eq. 67, it follows that

$$I_P(R) < I_P(r) \quad (71)$$

which immediately reduces to

$$R < r \quad (72)$$

and thus $r > D$, which holds over time $[t_o, \infty)$, guaranteeing SAI.

It can be shown that this condition is equivalent to finding a bounding zero-velocity surface between the two bodies.

7 Particular Solutions

While the general full 2-body problem cannot be solved in closed form, there are a number of solutions which can be found. In this chapter we only focus on relative equilibria for the system. These relative equilibrium solutions are particular solutions that satisfy the equations of motion. They do not provide a more solution to the equations of motion in terms of initial conditions, however given a relative equilibrium it is always possible to construct local solutions in phase space that approximate the solution.

In the following the conditions for a relative equilibrium are stated. Following this, specific applications to a variety of approximations to the general 2-body problem are given.

7.1 Relative Equilibrium and Minimum Energy Configurations

To find relative equilibria we can directly use the minimum energy function defined previously. An essential theorem due to Smale can be stated as follows:

Theorem 3 *A system is in relative equilibrium if and only if $\delta\mathcal{E} \geq 0$ for all allowable variations in the configuration of the system. If the function is positive definite at a relative equilibrium, the relative equilibrium is absolutely stable. Otherwise, it is not stable under the dissipation of energy.*

Proof 3 *The proof can be found in Smale's work. As regards the stability, if the energy function is indefinite or negative definite at a relative equilibrium it has well defined pathways by which it can diverge from the relative equilibrium if energy is dissipated. If it is positive definite it cannot lose energy and is locally trapped at that configuration.*

7.2 Relative Equilibrium Conditions

In the absence of constraints on the internal degrees of freedom of the system, conditions for a relative equilibrium are found by finding the conditions under which the minimum energy function is stationary with respect to the relative position, relative orientation, and the placement of the angular momentum vector relative to the system configuration. Taking the variation $\delta\mathcal{E}$ under the assumption that all parameters are independent yields the conditions:

$$\mathcal{E}_{\mathbf{r}} = 0 \quad (73)$$

$$\mathcal{E}_{\boldsymbol{\theta}} = 0 \quad (74)$$

$$\mathcal{E}_{\hat{\mathbf{H}}} = 0 \quad (75)$$

Even though these statements are independent, the body's internal configuration is coupled through these different conditions.

First consider the condition on the placement of the angular momentum vector. The variation only involves one term in the function, and thus has a simple form

$$\mathcal{E}_{\hat{\mathbf{H}}} = -\frac{H^2}{I_H^2} \hat{\mathbf{H}} \cdot \mathbf{I} \cdot \delta\hat{\mathbf{H}} \quad (76)$$

We note that the variation $\delta\hat{\mathbf{H}}$ is constrained to only occur normal to the vector $\hat{\mathbf{H}}$. From this we note that for the variation to be arbitrarily zero the term $\hat{\mathbf{H}} \cdot \mathbf{I}$ must be parallel to the direction $\hat{\mathbf{H}}$. But this just means that at a relative equilibrium the angular momentum vector must lie along a principal axis of the system inertia (which is a function of both the relative position and attitude). Further, if only interested in energetically stable relative equilibria, one would immediately choose the maximum moment of inertia axis of the dyadic $\mathbf{I}$, as only this one will be positive definite for a given system. Thus, given $\mathbf{r}$ and $\mathbf{A}$, there exists a rule for choosing the angular momentum vector orientation relative to the system, eliminating it from the solution process. Finally, knowing that the system must spin about a principal axis of inertia, this means that $\hat{\mathbf{H}}$ points along the rotation axis of the relative equilibrium.

Next consider the position variation term

$$\mathcal{E}_{\mathbf{r}} = -\frac{M_1 M_2}{M_1 + M_2} \frac{H^2}{I_H^2} \mathbf{r} \cdot [\mathbf{U} - \hat{\mathbf{H}} \hat{\mathbf{H}}] + U_{\mathbf{r}} \quad (77)$$

This term must be identically equal to zero. First assume that the bodies are not touching, the contact case will be considered later. Rearranging the defining equation yields

$$\mathbf{r} = (\mathbf{r} \cdot \hat{\mathbf{H}}) \hat{\mathbf{H}} + \frac{M_1 + M_2}{M_1 M_2} \frac{I_H^2}{H^2} U_{\mathbf{r}} \quad (78)$$

Thus the relative position between the bodies must lie in the plane defined by the gravitational force between the bodies and the rotation axis of the system.

Finally, taking the attitude variation term only involves the relative orientation between the bodies, which is uniquely accounted for in the system inertia by the body 2 inertia dyadic and the moment that body 1 exerts on body 2. This term must also be uniformly zero and can be reduced to

$$\mathcal{E}_\theta = -\frac{H^2}{I_H^2} \hat{\mathbf{H}} \cdot \mathbf{A} \cdot \mathbf{I}_2 \cdot \mathbf{A}^T \times \hat{\mathbf{H}} + U_\theta \quad (79)$$

Note that at a relative equilibrium the following condition holds, $H/I_H = \Omega$, the total spin rate of the system, and $\mathbf{\Omega} = \Omega \hat{\mathbf{H}}$. Thus, these conditions just devolve to the condition $\mathbf{I}_2 \cdot \dot{\mathbf{\Omega}} = 0$ from the equations of rotational motion for body 2. It is significant to note that this equilibrium, along with the other two, can be used to show that the analogous condition for body 1 is identically satisfied at the relative equilibrium.

Finally, the case of resting equilibrium must be considered. In this case, the two bodies touch and, under our smooth, convex shape hypothesis, at every point they touch the surfaces are mutually normal. The distance between the bodies is constrained by our touching limit $d(\hat{\mathbf{r}}, \mathbf{A})$. The remaining five degrees of freedom are still active, however. To find relative, resting equilibria one can still take variations for the relative location and attitude. These operations are similar to those above, with the addition of a fixed distance between the mass centers. The main difference lies in the contact constraint, stated as $\delta \mathcal{E} \geq 0$ for allowable variations $\delta d \geq 0$. Essentially, if all the other conditions are satisfied, this final condition corresponds to whether the total force between the bodies is compressive, meaning that work must be performed to separate the bodies. Then, with the remainder of other conditions, this implies that the bodies will be held together under their mutual gravitation. If, however, $\delta \mathcal{E} < 0$ under the same constraint, this means that work can be liberated by moving the bodies apart, or that the net force between them is in tension and, unless held together, will lose contact and not sustain the resting equilibrium.

In the following we move from the most restrictive assumption to less restrictive assumptions, showing how our results become significantly different as we move from the classical Kepler problem to the finite density problems of the Full Two Body Problem.

7.3 Point Mass Bodies

For the point mass case we can show that there is only one relative equilibrium. Starting from Eqn. 78 and inserting the gradient of the point mass potential we find

$$\mathbf{r} = (\mathbf{r} \cdot \hat{\mathbf{H}}) \hat{\mathbf{H}} + \frac{I_H^2}{H^2} \frac{\mathcal{G}(M_1 + M_2)}{r^3} \mathbf{r} \quad (80)$$

For the point mass case we note that $I_H = \frac{M_1 M_2}{M_1 + M_2} r^2$ and $H = \frac{M_1 M_2}{M_1 + M_2} \mathbf{r} \times \mathbf{v}$. Thus, the angular momentum vector is perpendicular to the radius vector and $\mathbf{r} \cdot \mathbf{H} = 0$. Identifying the two sides for any vector $\mathbf{r}$ we find

$$1 = \frac{I_H^2}{H^2} \frac{\mathcal{G}(M_1 + M_2)}{r^3} \quad (81)$$

For the point mass problem recall that $\mathbf{r} \times \mathbf{v} = r^2 \Omega \hat{\mathbf{H}}$. Then we can note that at a relative equilibrium $H = I_H \Omega$, where Ω is the system spin rate. Thus we find the condition for a relative equilibrium for the 2 point mass problem:

$$\Omega^2 = \frac{\mathcal{G}(M_1 + M_2)}{r^3} \quad (82)$$

which is just the rotation rate for a mutual circular orbit.

An alternate approach is to evaluate the minimum energy function for this condition and directly find the stationary points. Under the restrictions the function reduces to

$$\mathcal{E} = \frac{H^2}{2 \frac{M_1 M_2}{M_1 + M_2} r^2} - \frac{\mathcal{G} M_1 M_2}{r} \quad (83)$$

$$= \mathcal{G} M_1 M_2 \left[\frac{H^2 (M_1 + M_2)}{2 M_1^2 M_2^2 \mathcal{G}} \frac{1}{r^2} - \frac{1}{r} \right] \quad (84)$$

Defining a normalized angular momentum $\bar{H}^2 = \mathcal{G} M_1^2 M_2^2 / (M_1 + M_2)$ we can plot this function for arbitrary values of $\bar{H} \geq 0$. Thus geometrically (or functionally if we solve for the stationary point of this curve and its second derivative) we see that a minimum energy configuration exists for all $\bar{H} > 0$, located at $r = \bar{H}^2$. For the case of $\bar{H} = 0$ we see that there is no minimum defined.

7.4 Finite Sphere Bodies

Now consider the case of finite density spheres, focusing on relative equilibrium, local minimum configurations and globally minimum configurations. This problem shows that extensions to finite density mass distributions fundamentally changes the energy structure of this problem and yields the non-intuitive result that there can be multiple circular orbits at a given angular momentum and that there can be unstable circular orbits. For generality in the following the two bodies have different masses, M_1 and M_2 , and diameters, d_1 and d_2 .

Theorem 4 *There are 0, 1 or 2 orbital relative equilibria for the 2-sphere problem as a function of angular momentum. When there are 2 orbital equilibria, one is energetically stable and the other is energetically unstable.*

Proof 4 *The potential and polar moments of inertia are:*

$$U = -\frac{\mathcal{G}M_1M_2}{r} \quad (85)$$

$$I_H = \frac{M_1M_2}{M_1 + M_2}r^2 + I_s \quad (86)$$

$$I_s = \frac{1}{10}(M_1d_1^2 + M_2d_2^2) \quad (87)$$

$$r \geq \frac{d_1 + d_2}{2} = D \quad (88)$$

Normalize this system by

$$\bar{r} = r/D \quad (89)$$

$$\bar{H} = \frac{H\sqrt{M_1 + M_2}}{\sqrt{\mathcal{G}D}M_1M_2} \quad (90)$$

$$\bar{I}_s = \frac{M_1 + M_2}{D^2M_1M_2}I_s \quad (91)$$

$$\bar{U} = \frac{UD}{\mathcal{G}M_1M_2} \quad (92)$$

In the following the $\overline{(-)}$ notation will be dropped with the assumption that all quantities are normalized as given above. With this normalization the range of allowable distances between the particles is defined by $r \geq 1$.

The minimum energy function is then

$$\mathcal{E} = \frac{H^2}{2(r^2 + I_s)} - \frac{1}{r} \quad (93)$$

This is a single degree of freedom function so relative equilibria are found by taking a variation with respect to r

$$\delta\mathcal{E} = \left[-\frac{H^2}{(r^2 + I_s)^2} + \frac{1}{r^3} \right] r\delta r \quad (94)$$

Setting the variation equal to zero yields an equation for r :

$$r^4 - H^2r^3 + 2I_sr^2 + I_s^2 = 0 \quad (95)$$

Applying the Descartes rule of signs the polynomial either has 2 or no solutions, a drastic change from the point-mass problem which always had a single solution¹. Of

¹If $I_s \rightarrow 0$ the single solution from the point-mass case is recovered.

particular interest is the point where the transition from zero to 2 solutions occurs, which will be a double root of this polynomial. Set the derivative to zero to find

$$(4r^2 - 3H^2r + 4I_s)r = 0 \quad (96)$$

Evaluating both equations simultaneously shows that the bifurcation occurs at

$$r^* = \sqrt{3I_s} \quad (97)$$

$$H^2 = \frac{16}{3\sqrt{3}}\sqrt{I_s} \quad (98)$$

Note that the minimum value of I_s occurs when $M_1 = M_2$ which leads to $d_1 = d_2$ under an equal density assumption, which leads to $I_s \geq 0.4$. Thus, $r^* \geq \sqrt{1.2} > 1$ for all mass values with constant density and the bifurcation structure for $N = 2$ is robust and always occurs when the bodies are separate from each other.

An alternative approach is to solve for H^2 as a function of r , yielding

$$H^2 = \frac{(r^2 + I_s)^2}{r^3} \quad (99)$$

This has a minimum point, corresponding to the bifurcation of the roots to the equation given above in Eqns. 97 and 98. For higher values of H the system will then have two roots, one to the right of r^* (the outer solution) and the other to the left (the inner solution). At the bifurcation point the system technically only has one solution.

The stability of each of these solutions is determined by inspecting the second variation of $\mathcal{E}$ with respect to r , evaluated at the relative equilibrium. Taking the variation and making the substitution for H^2 from Eqn. 99 yields

$$\delta^2\mathcal{E} = \left[\frac{4}{(r^2 + I_s)} - \frac{3}{r^2} \right] \frac{(\delta r)^2}{r} \quad (100)$$

Checking for when $\delta^2\mathcal{E} > 0$ yields the condition $r^2 > 3I_s$. Thus, the outer solution will always be stable while the inner solution will always be unstable, with the relative equilibria occurring at a minimum and maximum of the minimum energy function, respectively. Figure 2 shows characteristic energy curves and the locus of equilibria for different levels of angular momentum for the spherical full 2-body problem. This should be contrasted with the energy function for the point mass 2-body problem which only has one relative equilibrium.

Theorem 5 *There are 0 or 1 unique resting relative equilibrium configurations as a function of angular momentum. When it exists, the resting equilibrium is energetically stable.*

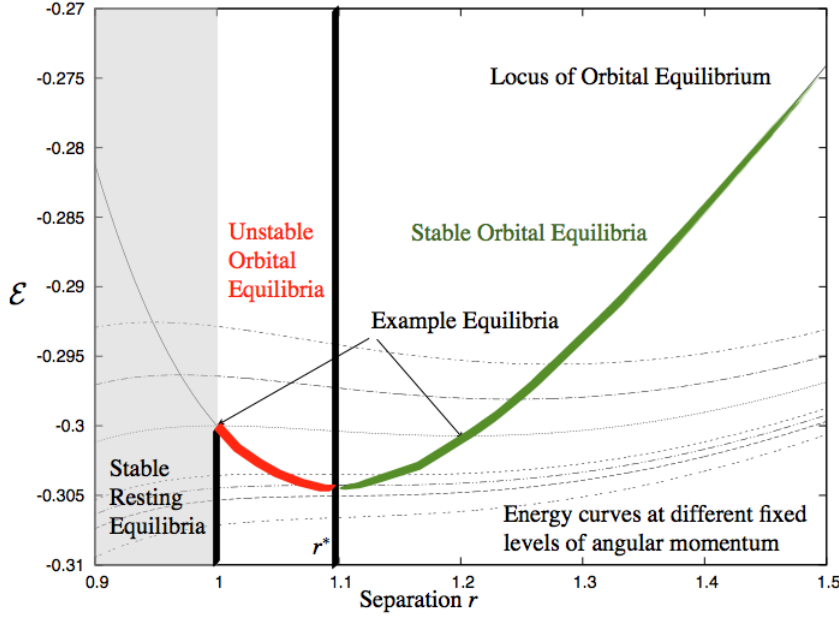


Figure 2: Locus of all equilibria across a range of angular momentum values. Plotted is the minimum energy function versus the system configuration, r , the distance between the two particles. This plot assumes equal masses and sizes of the two particles. For clarity, equilibrium solutions are shown below the physical limit $r \geq 1$.

Proof 5 *There is only a single unique contact configuration for the $N = 2$ case, with the two bodies touching with their centers of mass a distance D apart. In normalized coordinates the minimum energy function is $\mathcal{E} = \frac{H^2}{2(r^2 + I_s)} - \frac{1}{r}$ where $r = 1$ signifies contact between the bodies. For a contact structure to be a resting equilibrium, any variations in its degrees of freedom must conform to the rigid body constraint (i.e., $\delta r \geq 0$ in this case) and must only cause an increase in the minimum energy function, $\delta \mathcal{E} \geq 0$. If a decrease in energy occurs, the system will trend away from the configuration along an allowable variation in the degree of freedom. Taking the variation in the minimum energy function at the contact configuration of $r = 1$ yields*

$$\delta \mathcal{E} = \left[1 - \frac{H^2}{(1 + I_s)^2} \right] \delta r \quad (101)$$

Since $\delta r \geq 0$ is the only variation allowed, the relative equilibrium will exist when $1 - \frac{H^2}{(1 + I_s)^2} \geq 0$. Thus if $H \leq 1 + I_s$ the static resting equilibrium will exist and if $H > 1 + I_s$ there will be no resting equilibrium.

When the resting equilibrium exists, any variation in the degree of freedom will only increase the energy, meaning that its variation is positive definite and this relative equilibrium is energetically stable.

The complete bifurcation chart of relative equilibria, minimum energy states, and global minimum energy states of the sphere restricted $N = 2$ full body problem as a function of angular momentum follows. The sequence is graphically represented in Figure 3.

$0 \leq H^2 < \frac{16}{3\sqrt{3}}\sqrt{I_s}$ The minimum energy configuration, and the only relative equilibrium, is for the two bodies to be resting on each other with $r = D$.

$\frac{16}{3\sqrt{3}}\sqrt{I_s} \leq H^2 < \frac{4\sqrt{I_s}(1+I_s)}{\sqrt{1+I_s+\sqrt{I_s}}}$ Two orbital relative equilibria bifurcate into existence at a radius $r^* = \sqrt{3I_s}$ and grow to larger and smaller values of r . The outer solution is at a local minimum of the energy, and thus is stable. The inner solution is at a local maximum of the energy, and thus is unstable. The global minimum energy configuration remains the contact solution.

$\frac{4\sqrt{I_s}(1+I_s)}{\sqrt{1+I_s+\sqrt{I_s}}} \leq H^2 < (1+I_s)^2$ The situation is the same as above, except now the outer orbital solution becomes the global minimum, and the resting solution becomes a local minimum.

$(1+I_s)^2 = H^2$ The inner orbital solution collides with the resting solution and they annihilate each other. This can be called the “fission” condition for the resting configuration, as it is the point beyond which the resting condition no longer exists and the particles that were resting on each other are forced to enter orbit about each other.

$(1+I_s)^2 < H^2$ Beyond the fission limit there is only one relative equilibrium, the outer orbital solution, which is the minimum energy configuration.

The $N = 2$ minimum energy transition diagram as a function of angular momentum is discontinuous and exhibits hysteresis. In Figure 3 the energy of the relative equilibria is graphed when they are stable as a function of angular momentum for equal-sized bodies (with qualitatively similar results holding across all values). Under increasing or decreasing angular momentum the configuration will remain in a locally stable relative equilibrium after the global minimum configuration has switched to a different configuration. Thus, when the local relative equilibrium becomes unstable or ceases to exist, there is a finite energy difference between the current system state and its minimum energy state. If angular momentum is increased beyond this point this directly implies a period of dynamical interaction during which energy dissipation may occur, eventually leading to the minimum energy state.

At fission the system has a corresponding energy of $\frac{1}{2}(I_s - 1)$ and thus at fission the energy can be positive if $I_s > 1$. For constant density spheres this will occur when the ratio between the masses is $< 0.204\dots$ or $> 4.902\dots$, assuming constant density. We recall that full body systems can escape if their total energy is positive. Thus, across

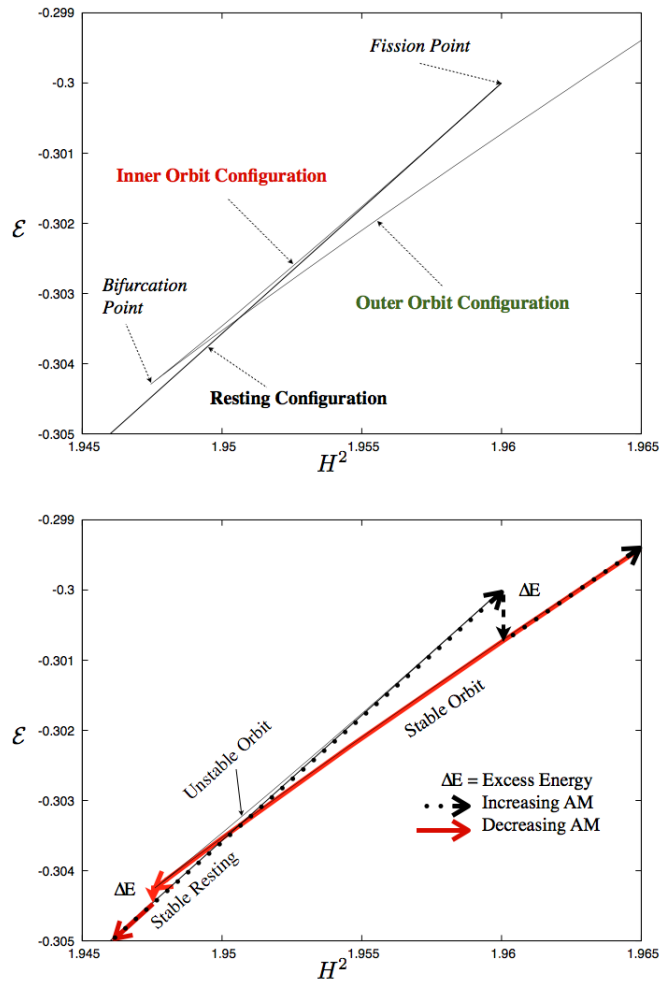


Figure 3: The energy-angular momentum diagram of relative equilibria (left) and the transition diagram for increasing and decreasing angular momentum (right) in the Full 2-body problem for equal sized bodies.

the range of different mass ratios it is even possible for the resulting fission system to eventually escape prior to sufficient energy dissipation for being trapped into the outer orbital configuration. Simulations have shown that this outcome for systems with mass ratios below the above limit will eventually escape. This same mass ratio is found in a class of natural bodies that are expected to have undergone fission in the past, showing the relevance of this result for natural systems.

7.5 Sphere-Restricted

For the relative equilibrium between a general ellipsoidal-like body and a sphere there is again a changing number of relative equilibria as a function of angular momentum. For low H there are 4 planar contact equilibrium and two out-of-plane polar contacts (see Fig. 4). As angular momentum increases there is an interval with two solutions along each of the planar principal axes, the inner one which is unstable and the outer one which can be stable for the intermediate axis and which is always stable for the minimum axis. The inner one collides with the ellipsoid, at which point the contact solution no longer exists. Subsequent to that, only the outer solutions exist but do so for all higher values of H . The number of planar relative equilibria will in general change with H from 4 to 8 to 12 to 8 to 4. Out of these, only 2 of the orbital relative equilibria will ever be energetically stable. Across the range of angular momentum values, it is possible for all of the contact equilibria to be stable, depending on the specific shape of the central body. Figure 4 graphically shows the various contact and orbital relative equilibria that can exist.

7.6 General Problem

For a discussion of relative equilibria in the un-approximated full body problem we will notionally use ellipsoid/ellipsoid systems again, as they have been well studied and understood. For two ellipsoids, where we can distinguish each separate configuration, there are 144 distinct ways for the bodies to be in relative orientation. Modulo the symmetries in the ellipsoid-ellipsoid problem, we can reduce these to 18 distinct situations (see Fig. 5 for four examples). For each configuration there are 3 distinct principle axes about which the system can be rotated, and two possible directions of rotation. For orbital equilibria, only two of the principle axes of each configuration will generate a relative equilibria, as rotation about the line connecting the centers of mass can never generate centripetal accelerations to balance the mutual attraction. Thus, for distinguishable ellipsoids there are $144 \times 2 \times 2 = 576$ different orbital relative equilibria configurations possible. Adding that there are generically two relative equilibria for each configuration, one inner and one outer, we see that there are up to 1152 different distinguishable relative equilibria between two triaxial ellipsoids. Modulo the symmetries in the ellipsoid-ellipsoid problem, this reduces to $18 \times 2 = 36$ different relative equilibrium configurations and 72 different relative equilibria. Out of all of these configurations, it is interesting to note that only four are likely to be energetically stable, those ones that have their minimum axes of inertia pointing towards each other and the entire system rotating about their maximum axes of inertia. Among those four there is only one that is distinct modulo symmetry between ellipsoid-ellipsoid problems.

In addition to the orbital relative equilibria, each distinct relative configuration can

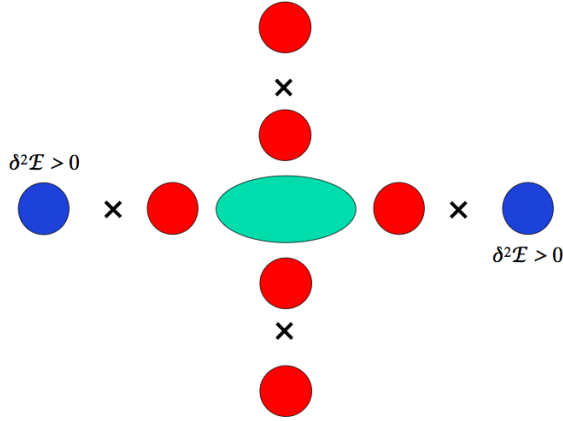
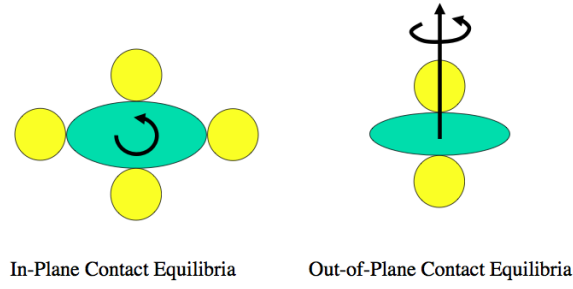


Figure 4: Graphical description of contact (left) and orbital (right) relative equilibria for the Sphere-Restricted problem. Blue are always energetically stable, yellow may be energetically stable, and red is never energetically stable.

be a contact equilibrium for a small enough angular momentum. For the distinguishable problem, each of these 144 configurations can be rotated about one of three axes in two separate directions, yielding $144 \times 3 \times 2 = 864$ contact relative equilibria. Modulo the symmetries in the ellipsoid-ellipsoid problem, we have $18 \times 3 = 54$ distinct situations. Out of these, there are several that can be minimum energy configurations as a function of the angular momentum.

8 Solution of the 2-Point Mass or 2-Sphere Problem

Finally, we consider the dynamics of two point mass or finite spheres as they move about each other. This particular problem can be solved exactly, yielding Kepler's solution for the 2-body problem.

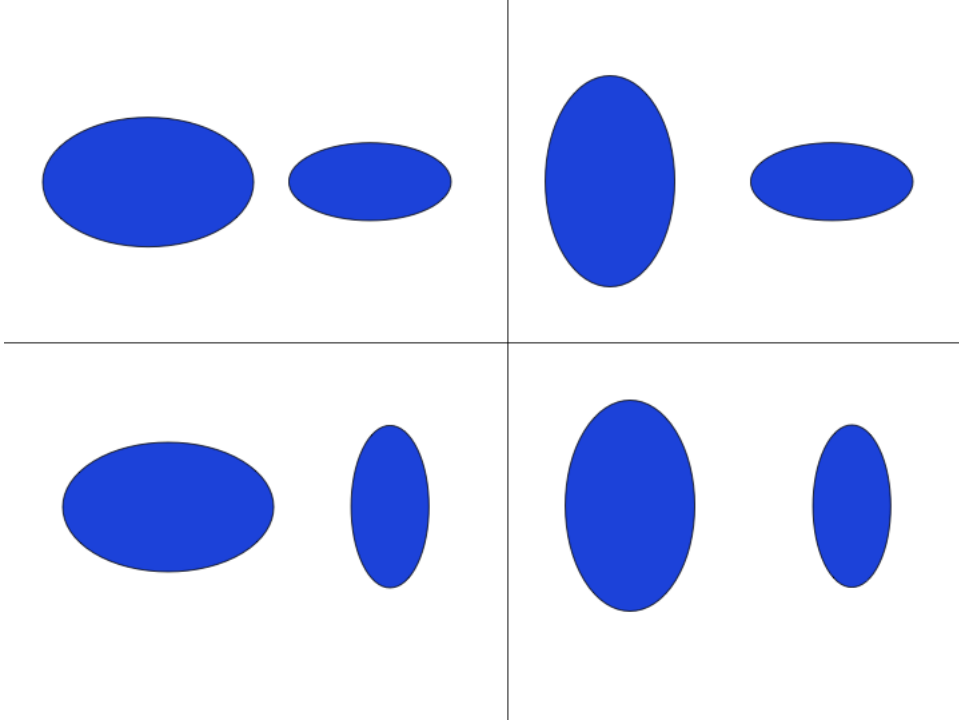


Figure 5: Graphical description of four distinct planar relative configurations for the ellipsoid-ellipsoid problem under symmetry. If the relative orientations are distinguishable then there are 16 possible orientations. For the planar case, each configuration can have up to three relative equilibria as a function of angular momentum, one resting and two orbital. Only the orbital configuration in the upper left corner can be energetically stable.

8.1 Equations of Motion

Following from Eqn. 43, we can state this problem as the relative motion of the two bodies

$$\ddot{\mathbf{r}} = -\frac{M_1 + M_2}{M_1 M_2} U \mathbf{r} \quad (102)$$

Usual practice is to replace the potential energy with the specific force potential for this problem

$$V(\mathbf{r}) = -\frac{M_1 + M_2}{M_1 M_2} U \quad (103)$$

$$= \frac{\mathcal{G}(M_1 + M_2)}{|\mathbf{r}|} \quad (104)$$

Then the equations of motion are more simply stated as

$$\ddot{\mathbf{r}} = V \mathbf{r} \quad (105)$$

where

$$V\mathbf{r} = -\frac{\mu}{r^3}\mathbf{r} \quad (106)$$

and we introduce the standard notation $\mu = \mathcal{G}(M_1 + M_2)$.

8.2 Classical Integrals of Motion

For the Kepler 2-body problem the integrals of angular momentum and energy only involve the relative motion of the two bodies, as the rotational dynamics of the bodies either don't exist (for the point mass case) or are completely decoupled from the orbital motion (for the sphere case). It is also customary to deal entirely with the specific angular momentum and energy, meaning that these quantities are normalized by the reduced mass of the system, $M_1 M_2 / (M_1 + M_2)$. We have already introduced this normalization above by defining the force potential $V(\mathbf{r})$.

With these normalizations the angular momentum and energy of the problem become

$$E' = \frac{1}{2}\mathbf{v} \cdot \mathbf{v} - V(\mathbf{r}) \quad (107)$$

$$\mathbf{h} = \mathbf{r} \times \mathbf{v} \quad (108)$$

where $\mathbf{v} = \dot{\mathbf{r}}$. Conservation of energy E' relies on the force potential V not being a function of time, while conservation of the angular momentum $\mathbf{h}$ relies on V corresponding to a central field, i.e., only a function of the distance between the two bodies. The simple two-body potential satisfies both of these conditions, as can be verified. In the following we will suppress the $'$ notation on the specific energy and just use E . We note that the specific angular momentum is now stated as a lower-case quantity.

The geometrical interpretation of the angular momentum integral can be split into two pieces, and is most easily noted by rewriting it as $\mathbf{h} = h\hat{\mathbf{h}}$, the product of the magnitude and the unit vector in which it points, both of which must be constant. Constancy of $\hat{\mathbf{h}}$ implies that the solution to the two body problem lies on a fixed plane perpendicular to $\hat{\mathbf{h}}$. This plane can be oriented relative to an inertial coordinate system by two angles, traditionally chosen as the inclination, i , which is the angle between $\hat{\mathbf{h}}$ and the $\hat{\mathbf{z}}$ axis of the coordinate frame, and the longitude of the ascending node Ω , which is the angle from the $\hat{\mathbf{x}}$ axis to the vector $\hat{\mathbf{z}} \times \hat{\mathbf{h}}$, which lies in the $\hat{\mathbf{x}} - \hat{\mathbf{y}}$ plane. The angular momentum unit vector can be specified using these two angles as:

$$\hat{\mathbf{h}} = \sin i \sin \Omega \hat{\mathbf{x}} - \sin i \cos \Omega \hat{\mathbf{y}} + \cos i \hat{\mathbf{z}} \quad (109)$$

The angular momentum being a constant of the motion, these two “orbital elements” are also constants.

The constancy of the orbit energy and angular momentum magnitude provide explicit relationships between the velocity and the position of the relative solution in the plane defined by $\hat{\mathbf{h}}$. If we define the angle between the position and velocity vector as $\pi/2 - \gamma$, where γ is commonly termed the flight path angle, then $h = rv \cos \gamma$. It is interesting to consider the projection of the velocity vector along the radius vector and perpendicular to it:

$$v_r = \mathbf{v} \cdot \hat{\mathbf{r}} \quad (110)$$

$$= v \sin \gamma \quad (111)$$

$$v_h = v \cos \gamma \quad (112)$$

$$= h/r \quad (113)$$

where v_r stands for the radial velocity and v_h stands for the horizontal velocity. If we introduce the angular rotation rate $\dot{\theta}$ of the position vector relative to inertial space, using an implied polar coordinate system in the orbit plane with center at body P_1 , and note the familiar kinematic relationship $v_h = r\dot{\theta}$, we find

$$h = r^2 \dot{\theta} \quad (114)$$

Next consider the orbit energy, E , and the constraints it places on the motion. There is a strong distinction between positive and negative energy solutions, with fundamentally different behavior resulting between the two cases. Consider a system with total negative energy, $E < 0$. Then from the energy equation it is simple to derive an upper bound on the distance between the two bodies. For the two body problem our specific argument rests on the fact that $\lim_{r \rightarrow \infty} V(\mathbf{r}) = 0$, $\partial V / \partial r < 0$ for all r , and $\lim_{r \rightarrow 0} V(\mathbf{r}) \rightarrow \infty$. Then, there exists a radius R_{max} such that $V(R) + E \leq 0$ for $R \geq R_{max}$. But $V(r) + E = \frac{1}{2}v^2 \geq 0$. Thus, $V(r) \geq V(R)$ which leads to $r \leq R_{max}$, an upper bound on the distance between the two bodies. When the system reaches this upper bound, the kinetic energy of the system is zero and the energy is completely contained in the potential energy of the system. For the case of positive energy there are no upper bounds on the motion, and for the point mass two body problem all solutions will eventually travel to arbitrarily large separations between the bodies.

Combining the angular momentum magnitude and the energy integrals we can establish a lower bound on the energy for a given angular momentum. Specifically we find that for a given angular momentum h , the energy must be greater than or equal to $E_{min} = -\frac{\mu^2}{2h^2}$. To show this start with the supposition

$$E = \frac{1}{2}v^2 - \frac{\mu}{r} \geq -\frac{\mu^2}{2h^2} \quad (115)$$

Removing the speed v from the above inequality with the substitution $v = h/(r \cos \gamma)$, we can find the quadratic equation

$$r^2 - \frac{2h^2}{\mu}r + \frac{h^4}{\mu^2 \cos^2 \gamma} \geq 0 \quad (116)$$

and its factorization

$$r = \frac{h^2}{\mu} \left[1 \pm \sqrt{1 - \frac{1}{\cos^2 \gamma}} \right] \quad (117)$$

Note that the radicand is less than or equal to 0, and thus that a physical value of radius r can never equal its root except when $\gamma = 0$ and $r = h^2/\mu$. At this value of γ and r , however, the energy is identically equal to E_{min} . For any other value of γ the energy is greater than E_{min} , establishing the result.

Continuing with the angular momentum and energy we can also find a lower bound on the radius. First, we assume that $h > 0$ and that a minimum separation distance between the bodies exists. Then, at a minimum distance we must have $d(\mathbf{r} \cdot \mathbf{r})/dt = 0$, or $\mathbf{r} \cdot \mathbf{v} = 0$, and thus the flight path angle $\gamma = 0$ at a minimum of the separation radius leading to $h = r_p v_p$. Eliminating v_p from between the energy and angular momentum then provides the result:

$$E = \frac{h^2}{2r_p^2} - \frac{\mu}{r_p} \quad (118)$$

which can be changed into the quadratic equation and its solution

$$r_p^2 + \frac{\mu}{E} r_p - \frac{h^2}{2E} = 0 \quad (119)$$

$$r_p = \frac{-\mu}{2E} \left[1 \pm \sqrt{1 + \frac{2Eh^2}{\mu^2}} \right] \quad (120)$$

Note that for a physical orbit the radicand must always be greater or equal to zero, given the lower limit on energy. Thus the solution always has at least one physical (i.e., real) root. If $E > 0$, then the radicand is always real and greater than 1, and the $-$ sign has a positive separation between the two bodies. If $E < 0$ we note that $E \geq E_{min}$ and the radicand is positive and less than 1. Thus in this case there are two real solutions, inspection indicates that one is a minimum and one a maximum. Thus the minimum radius of the system can be denoted as:

$$r \geq r_p \quad (121)$$

$$r_p = \frac{\mu}{2E} \left[\sqrt{1 + \frac{2Eh^2}{\mu^2}} - 1 \right] \quad (122)$$

Conversely, whether there is a maximum radius depends on the energy of the system, leading to the result $r \leq r_a$ with

$$r_a = \begin{cases} \frac{-\mu}{2E} \left[1 + \sqrt{1 + \frac{2Eh^2}{\mu^2}} \right] & E < 0 \\ \infty & E \geq 0 \end{cases} \quad (123)$$

8.3 Additional Integrals

While strong constraints can be placed on the planar motion for the two body motion, the solution is not completely described as of yet. The general full 2-body problem has the same integrals of motion discussed to this point, translational momentum, angular momentum, and energy, resulting in 10 separate integrals of motion that can be used to reduce the dimension of the system and place constraints on its motion. In the point-mass two-body problem we note that the system can be reduced to a one degree of freedom system by use of the Clairaut solution substitution. We will proceed in a different direction. First note the existence of an additional integral of motion which is only defined for the two-body problem, the Laplace vector or equivalently the eccentricity vector. Second note Jacobi's final integral theorem, which states that any system that has been reduced to a one-dimensional dynamical system can always be solved by quadratures.

The Laplace vector can be formed by the combination:

$$\mathbf{b} = \mathbf{v} \times \mathbf{h} - \mu \hat{\mathbf{r}} \quad (124)$$

while the eccentricity vector is just:

$$\mathbf{e} = \frac{\mathbf{b}}{\mu} \quad (125)$$

$$= \frac{1}{\mu} \mathbf{v} \times \mathbf{h} - \hat{\mathbf{r}} \quad (126)$$

It can be shown that $\dot{\mathbf{e}} = 0$ by direct substitution of the previous integrals of motion and the equations of motion. Specifically,

$$\dot{\mathbf{e}} = -\frac{1}{r^3} \mathbf{r} \times \mathbf{h} - \frac{1}{r} [\mathbf{I} - \hat{\mathbf{r}}\hat{\mathbf{r}}] \cdot \mathbf{v} \quad (127)$$

Note that $\mathbf{r} \times \mathbf{h} = \tilde{\mathbf{r}} \cdot \tilde{\mathbf{r}} \cdot \mathbf{v} = r^2 [\hat{\mathbf{r}}\hat{\mathbf{r}} - \mathbf{U}] \cdot \mathbf{v}$, showing that these terms are equal and opposite, leading to $\dot{\mathbf{e}} = 0$.

The eccentricity vector only has one additional item of information apart from the classical integrals. As a vector, it lies in the orbit plane of motion and its magnitude equals $e^2 = 1 + 2h^2E/\mu^2$, neither of which provides new information. The additional information that $\mathbf{e}$ provides is related to its orientation in the plane of motion, which provides an inertially fixed reference point for the relative motion between the bodies. Denote the angle between $\mathbf{e}$ and the node vector defined by $\hat{\mathbf{z}} \times \mathbf{h}$ as ω , called the argument of periapsis and another orbital element.

To solve for the trajectory of the bodies take the dot product of the constant vector $\mathbf{e}$ with the radius vector $\mathbf{r}$ to find:

$$\mathbf{r} \cdot \mathbf{e} = \frac{h^2}{\mu} - r \quad (128)$$

$$re \cos f = \frac{h^2}{\mu} - r \quad (129)$$

and where the angle between $\mathbf{r}$ and $\mathbf{e}$ is denoted as f and called the true anomaly. This equation can be explicitly solved for the distance between the two bodies to find:

$$r = \frac{h^2/\mu}{1 + e \cos f} \quad (130)$$

which gives us the relative distance between the two bodies as a function of the angle f measured from the $\mathbf{e}$ vector, which in turn is specified by the angle ω measured in the orbital plane from the node crossing (see Fig. 6 for a description of the geometry). This simple relationship captures Kepler's 1st law of planetary motion.

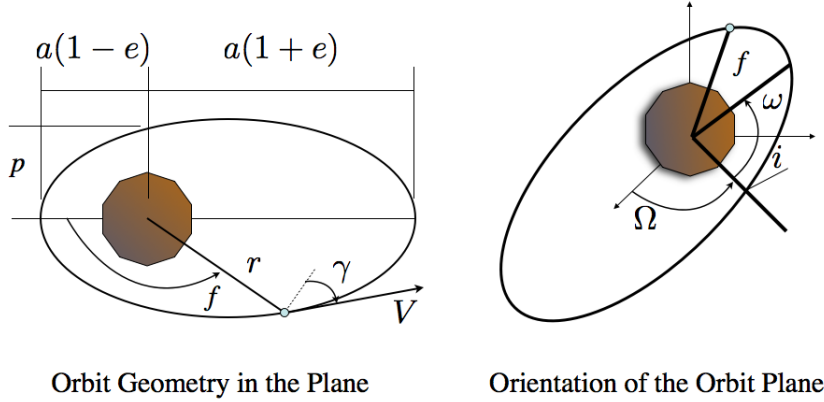


Figure 6: Geometric definition of the orbit elements.

The eccentricity, e , equals $\sqrt{1 + 2h^2E/\mu^2}$. Due to the minimum energy value constraint $e \geq 0$, with $e = 0$ occurring at the minimum energy for a given angular momentum (equivalent to the relative equilibrium found earlier). Thus, the eccentricity is a measure of the excess energy that an orbit has. If $E \geq 0$ then $e \geq 1$, which yields a fundamental change in the radius formula as now the denominator can disappear and, apparently $r \rightarrow \infty$. Denote the value h^2/μ as the orbit parameter, p , a sole function of the angular momentum of the orbit with units of length. With this new notation the trajectory solution for the two body problem takes on its standard form:

$$r = \frac{p}{1 + e \cos f} \quad (131)$$

Using this relation it is now possible to relate the speed and flight path angle as functions of the true anomaly f . From the energy equation the speed can be evaluated and from the angular momentum the flight-path angle can be evaluated

$$v = \sqrt{\frac{\mu}{p} (1 + 2e \cos f + e^2)} \quad (132)$$

$$\tan \gamma = \frac{e \sin f}{1 + e \cos f} \quad (133)$$

Again, these formula are valid for all energy and non-zero angular momentum. The only constraint which occurs is when $e \geq 1$, as then $1 + e \cos f = 0$ can occur at two particular values of true anomaly, $f = \pm f_\infty$ where $f_\infty = \arccos(-1/e)$, at which point the radius formula is undefined. Thus $f \in (-f_\infty, f_\infty)$. The additional quantities p , e and ω are all considered to be “classical” orbital elements in addition to the previously defined inclination i and longitude of the ascending node Ω . Note that another, alternate, orbit element is usually used known as the semi-major axis, a . This is a direct function of the energy only and can be computed as $a = -\frac{\mu}{2E}$, or alternately $a = p/(1 - e^2)$. Thus when $E < 0$, $a > 0$ and vice-versa. When $E = 0$ the quantity a is undefined.

For an orbit with zero energy, a parabolic orbit, $f_\infty = \pi$. The radius, velocity and flight path angle all take on special forms, arising from trigonometric reductions of the formula:

$$r = \frac{p}{2 \cos^2(f/2)} \quad (134)$$

$$v = 2 \sqrt{\frac{\mu}{p}} \cos(f/2) \quad (135)$$

$$\tan \gamma = \tan(f/2) \quad (136)$$

In this case as the true anomaly approaches $\pm f_\infty$ the speed approaches zero and the flight path angle approaches $\pi/2$.

For an orbit with positive energy, a hyperbolic orbit, f_∞ lies in the interval $(\pi, \pi/2)$, with its value approaching π as $e \rightarrow 1$ and $\pi/2$ as $e \rightarrow \infty$. The latter situation occurs when $\mu \rightarrow 0$, which in fact mimics the situation when a spacecraft has a flyby of a small asteroid. While the form of the radius, speed and flight path angle formulae do not change for this case, their limiting values do. Now as true anomaly approaches $\pm f_\infty$ the following limits occur, $v \rightarrow v_\infty$ and $\gamma \rightarrow \gamma_\infty$, where

$$v_\infty = \sqrt{\frac{\mu(e^2 - 1)}{p}} \quad (137)$$

$$= \sqrt{\frac{\mu}{-a}} \quad (138)$$

$$\gamma_\infty = \frac{\pi}{2} \quad (139)$$

For $E > 0$ recall that $e > 1$ and $a < 0$, thus yielding positive quantities within all of the relevant square roots.

With the results found to this point it is possible to define the 3-dimensional motion of the 2-body solution. Using the derived directions of $\hat{\mathbf{h}}$ and $\hat{\mathbf{e}}$ the vector $\hat{\mathbf{e}}_\perp = \hat{\mathbf{h}} \times \hat{\mathbf{e}}$ can be defined, and using $\hat{\mathbf{r}}$ can also define $\hat{\boldsymbol{\theta}} = \hat{\mathbf{h}} \times \hat{\mathbf{r}}$. Then, by applying these definitions the full position and velocity vector describing the orbit is:

$$\mathbf{r} = r [\cos f \hat{\mathbf{e}} + \sin f \hat{\mathbf{e}}_\perp] \quad (140)$$

$$\mathbf{v} = \frac{h}{r} [\hat{\boldsymbol{\theta}} + \tan \gamma \hat{\mathbf{r}}] \quad (141)$$

In these expressions r , v , γ , $\hat{\mathbf{r}}$, and $\hat{\boldsymbol{\theta}}$ all vary with true anomaly, with the appropriate limits for the cases of $E \geq 0$.

8.4 Motion in Time

What has been defined is the path, or trajectory, that a solution to the 2-body problem takes in position and velocity space, also called phase space. The link between the true anomaly and time has not been developed yet. Also, the reader can note that we've only defined 5 independent integrals to this point. These are the angular momentum $\mathbf{h}$, energy E , and eccentricity vector $\mathbf{e}$, which have 5 independent quantities that can be expressed in terms of the classical orbital elements. These are the orientation of the orbit plane, inclination i and longitude of ascending node Ω , orientation of the trajectory in the orbit plane, the argument of periapsis ω , and the shape and size of the orbit, the eccentricity e and either the semi-major axis a or the orbit parameter p .

What is lacking is the integral that locates where the solution is located within the trajectory at a given time. This final integral always exists due to a theorem by Jacobi. This theorem states that a 1-dimensional ordinary differential equation $\dot{x} = f(x)$ can always be solved by quadratures as

$$t - \tau_o = \int_{x_o}^x \frac{dx'}{f(x')} \quad (142)$$

with the epoch time τ_o serving as the final integral of motion. This result can either hold globally or locally between zeros of $f(x)$. In the two body problem it is easiest to find it by noting a simple relationship between the time rate of change of the true anomaly and the angular momentum magnitude.

Recall the angular momentum magnitude formula, $h = rv \cos \gamma$, and its alternate statement $h = r^2 \dot{\theta}$, where $\dot{\theta}$ is the time rate of change of the angular location of the

solution. For the two-body problem this will equal the time rate of change of the true anomaly, $\dot{f}$. The quantity $r^2\dot{\theta}$ equals twice the rate at which the radius vector sweeps out area, which is the classical statement of Kepler's 2nd law. Solving for $\dot{f}$ and evaluating r yields

$$\dot{f} = \sqrt{\frac{\mu}{p^3}}(1 + e \cos f)^2 \quad (143)$$

This differential equation can be solved by separation of variables, putting it into the form

$$\sqrt{\frac{\mu}{p^3}}(t - \tau_o) = \int_{f_o}^f \frac{df'}{(1 + e \cos f')^2} \quad (144)$$

We have the freedom to choose one of the initial conditions τ_o or f_o , and following convention choose $f_o = 0$, at periapsis, with τ_o then signifying the time at which the orbit was at periapsis, which then becomes the final integral of motion.

First consider the case $E < 0$ and $0 < e < 1$. The quadrature can be carried out for this case, yielding

$$\begin{aligned} \sqrt{\frac{\mu}{p^3}}(t - \tau_o) = & \left. \frac{-e}{1 - e^2} \frac{\sin f'}{1 + e \cos f'} \right|_{f_o}^f \\ & + \frac{2}{(1 - e^2)^{3/2}} \arctan \left\{ \sqrt{\frac{1 - e}{1 + e}} \tan \frac{1}{2} f' \right\} \Big|_{f_o}^f \end{aligned} \quad (145)$$

Evaluating this integral from $f = 0$ to 2π defines the period of the orbit, T , which takes on a particularly simple form when stated in terms of the semi-major axis:

$$T = \frac{2\pi a^{3/2}}{\sqrt{\mu}} \quad (146)$$

and is Kepler's 3rd law. Thus the period is only a function of the orbit energy. From this the mean motion can be defined as $n = \sqrt{\mu/a^3} = 2\pi/T$, as this is by definition the average rate at which the orbit is traversed.

The expression in Eq. 145 is not convenient to work with, leading to the definition of the eccentric anomaly, E . The arc-tangent function will define an angle, leading to the natural definition

$$\tan E/2 = \sqrt{\frac{1 - e}{1 + e}} \tan f/2 \quad (147)$$

Since this is a half-angle formula it yields angles resolved over the full interval $[0, 2\pi)$. Corresponding formula, and their inverses, for $\cos E$ and $\sin E$ in terms of $\cos f$ and $\sin f$

are stated below.

$$\cos E = \frac{e + \cos f}{1 + e \cos f} \quad (148)$$

$$\sin E = \frac{\sqrt{1 - e^2} \sin f}{1 + e \cos f} \quad (149)$$

$$\cos f = \frac{\cos E - e}{1 - e \cos E} \quad (150)$$

$$\sin f = \frac{\sqrt{1 - e^2} \sin E}{1 - e \cos E} \quad (151)$$

Using these definitions gives us the standard form of Eq. 145, which is usually called Kepler's equation:

$$n(t - \tau_o) = E - e \sin E \quad (152)$$

The mean anomaly is defined as $M = nt$, an additional angle. Then the final integral of motion can also be restated as $M_o = n\tau_o$. This gives the usual definition of Kepler's Equation

$$M - M_o = E - e \sin E \quad (153)$$

Given a true anomaly the mean anomaly can be directly computed, first transforming into the eccentric anomaly and then using Kepler's equation. Evaluation of the true anomaly given a mean anomaly is not as straightforward, as Kepler's equation must be solved to find the eccentric anomaly that corresponds to a give mean anomaly. Despite the simplicity of this equation, it cannot be solved in closed form given its transcendental nature. Significant research and analysis of this equation has occurred historically, and indeed still occurs today.

Equivalent quadratures and definitions can be carried out for the case $e = 1$ and $e > 1$, for parabolic and hyperbolic orbits. These relations are needed when modeling the flyby of a small body relative to a planet or of a spacecraft relative to a small body. As mentioned above, the other trajectory relations still hold, the only modification needed is to Kepler's equation. Going back to the fundamental quadrature,

$$\sqrt{\frac{\mu}{p^3}}(t - \tau_o) = \int_0^f \frac{df}{(1 + e \cos f)^2}$$

when $e > 1$ the result is

$$\begin{aligned} \sqrt{\frac{\mu}{p^3}}(t - \tau_o) = & \frac{e}{e^2 - 1} \frac{\sin f'}{1 + e \cos f'} \Big|_{f_o}^f \\ & - \frac{2}{(e^2 - 1)^{3/2}} \operatorname{arctanh} \left\{ \sqrt{\frac{e - 1}{e + 1}} \tan \frac{1}{2} f' \right\} \Big|_{f_o}^f \end{aligned} \quad (154)$$

Following the same approach for the elliptic case, define the Hyperbolic Anomaly, F , as

$$\tanh F/2 = \sqrt{\frac{e-1}{e+1}} \tan f/2 \quad (155)$$

with similar relationships between $\cos f$ and $\sin f$ and $\cosh F$ and $\sinh F$. Applying these substitutions yields the hyperbolic version of Kepler's equation:

$$\sqrt{\frac{\mu}{|a|^3}}(t - \tau_o) = e \sinh F - F \quad (156)$$

For parabolic orbits, the quadrature takes on a simpler form and yields a direct relationship between time and true anomaly:

$$\sqrt{\frac{\mu}{p^3}}(t - \tau_o) = \frac{1}{2} \tan\left(\frac{f}{2}\right) \left[1 + \frac{1}{3} \tan^2\left(\frac{f}{2}\right)\right] \quad (157)$$

where p is generally well defined (as it is computed from the angular momentum) even if $e = 1$.

8.5 Special Case of Zero Angular Momentum

Finally, note that for orbits with zero angular momentum, $p = 0$, the entire analysis must be redone given the singular definition of the true anomaly in this case. If the system has zero angular momentum this implies that $\mathbf{h} = \mathbf{r} \times \dot{\mathbf{r}} = \mathbf{0}$, which for non-zero $\mathbf{r}$ and $\dot{\mathbf{r}}$ only occurs if these two vectors are parallel to each other. Due to the constancy of the angular momentum, this then implies that the relative velocity and position of the system remain parallel or anti-parallel. Thus, motion is confined to a line in space. The eccentricity vector definition becomes $\mathbf{e} = -\hat{\mathbf{r}}$. Thus, the eccentricity $e = |\mathbf{e}| = 1$ for all cases of motion, not just for parabolic orbits. The periapsis vector is defined in the opposite direction of the unit position vector. The energy is still well defined with hyperbolic, parabolic and elliptic orbits having a total energy $E > 0$, $E = 0$, and $E < 0$, generically. Due to this the semi-major axis is still well defined as $a = -\frac{\mu}{2E}$.

With these definitions, we find that the periapsis radius of the system is always equal to zero, implying that the point masses impact with each other every periapsis passage. The apoapsis radius for an elliptic orbit will equal $2a$, and will be infinite for a parabolic and hyperbolic orbit. To orient the trajectory we still require two “elements,” but now they give the orientation of the trajectory line in space. Finally, due to the periapsis being at a zero radius the argument of periapsis is not well defined, although the direction of periapsis is $-\hat{\mathbf{r}}$.

In the following we discuss the explicit solution for orbital motion in this case, dividing the discussion again into the three cases for negative energy, zero energy and positive energy. We first discuss the energy integral and the general formulation of the problem solution.

8.5.1 Energy Integral and the General Solution

Forming the energy integral, $E = \frac{1}{2}v^2 - \frac{\mu}{r}$, we note that since $\mathbf{v} = \dot{\mathbf{r}}$ is parallel to $\mathbf{r}$, that $v = \dot{r}$, just the time derivative of the scalar radius r . Thus the energy integral is more properly stated as

$$E = \frac{1}{2} \left(\frac{dr}{dt} \right)^2 - \frac{\mu}{r} \quad (158)$$

and we can explicitly solve for the time rate of change of the radius to find

$$\frac{dr}{dt} = \pm \sqrt{2E + \frac{2\mu}{r}} \quad (159)$$

Separation of variables can be applied to the two differential quantities to yield the following form

$$dt = \pm \sqrt{\frac{r}{2\mu + 2Er}} dr \quad (160)$$

This is easily reduced to quadratures to provide a solution that gives time as a function of radius, choosing the passage through a radius of 0 (i.e., periapsis) to be the point from which motion is measured

$$t - \tau_o = \pm \int_0^r \sqrt{\frac{r'}{2\mu + 2Er'}} dr' \quad (161)$$

The $\pm$ corresponds to whether the relative solution is heading away from periapsis or towards periapsis. Inversion of this integral then represents the desired solution. This inversion can be performed, although the three different cases of $E < 0$, $E = 0$ and $E > 0$ must be evaluated separately. We consider the parabolic case first, as this is the simplest, and then consider the elliptic and hyperbolic cases.

8.5.2 Parabolic Solution for the Zero Angular Momentum Case

If the orbit is parabolic, then $E = 0$ and the solution reduces to the simplified version

$$t - \tau_o = \pm \int_0^r \sqrt{\frac{r'}{2\mu}} dr' \quad (162)$$

$$= \pm \frac{2}{3} \frac{r^{3/2}}{\sqrt{2\mu}} \quad (163)$$

Inversion of this integral is trivial and yields the explicit solution

$$r = \left(\frac{9\mu}{2} \right)^{1/3} |t - \tau_o|^{2/3} \quad (164)$$

where we note that the absolute magnitude of $t - \tau_o$ is taken. it is interesting to note that this equation for the radius does not have a singularity at periapsis, although the velocity of the orbit is unbounded at that point. Similar results hold for the other cases as well.

8.5.3 Elliptic Solution for the Zero Angular Momentum Case

For an elliptic orbit we use the traditional form of the energy integral $E = -\frac{\mu}{2a}$ and rewrite the general solution as

$$t - \tau_o = \frac{\pm 1}{\sqrt{2\mu}} \int_0^r \sqrt{\frac{r'}{1 - r'/(2a)}} dr' \quad (165)$$

This integral can be evaluated in closed form and yields

$$t - \tau_o = \mp \sqrt{\frac{a}{\mu}} \left[\sqrt{r(2a - r)} + 2a \arctan \left\{ \sqrt{\frac{2a - r}{r}} \right\} \right] \pm \pi \sqrt{\frac{a^3}{\mu}} \quad (166)$$

The inversion of the integral for radius as a function of time is not obvious, but can be simplified by the substitution of the radius as a function of the eccentric anomaly (now independent of the true anomaly). It is an exercise to show that, given this substitution, the above equation simplifies to Kepler's equation

$$r = a(1 - \cos E) \quad (167)$$

$$\sqrt{\frac{\mu}{a^3}}(t - \tau_o) = \pm (E - \sin(E)) \quad (168)$$

8.5.4 Hyperbolic Solution for the Zero Angular Momentum Case

Finally for a hyperbolic orbit we use the traditional form of the energy integral $E = \frac{\mu}{2|a|}$, noting that the semi-major axis is now negative, and rewrite the general solution as

$$t - \tau_o = \frac{\pm 1}{\sqrt{2\mu}} \int_0^r \sqrt{\frac{r'}{1 + r'/(2|a|)}} dr' \quad (169)$$

This integral can be evaluated in closed form and involves hyperbolic functions in general. A similar use of the Hyperbolic Eccentricity again yields the solution as a function of time, through Kepler’s equation again. Specifically it can be shown to result in

$$r = |a| (\cosh F - 1) \quad (170)$$

$$\sqrt{\frac{\mu}{|a|^3}}(t - \tau_o) = \pm (\sinh F - F) \quad (171)$$

9 Summary

The general statement of the gravitational two-body problem is given. Starting from this formulation, several fundamental results are derived and stated for the dynamical evolution of this system. The range of different approximations to this problem are reviewed and their implications are discussed. A particular focus on the relative equilibria in this problem is introduced, as these correspond to particular solutions of the problem, and some differences between the conventional point mass two-body problem and the finite density two-body problem are outlined. In particular, the stability of contact and orbital relative equilibria are discussed. Finally, the exact solution to the point-mass two-body problem is derived in detail.

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Glossary

Amended Potential: The total energy evaluated at a given level of angular momentum.

Energetic Stability: When a system is at its minimum energy configuration, meaning that it cannot dissipate any additional energy without losing angular momentum as well.

Full 2-Body Problem: The dynamical motion of two bodies, possibly distributed and non-spherical, as they evolve under mutual gravitational attraction.

Hill Stability: When the relative positions in a system are bounded for all future time.

Minimum Energy Configuration: A relative equilibrium that is also the minimum energy state for a given level of system angular momentum.

Minimum Energy Function: An alternate name for the amended potential, serves as a lower bound on the total system energy as a function of angular momentum and potential energy.

Particular Solutions: A closed-form solution to the gravitational N -body problem that only exists for a specific set of initial conditions.

Relative Equilibrium: When the members of a system are at rest relative to each other, even though the entire system may not be at rest relative to an inertial frame.

Resting Configuration: A relative equilibrium where at least some of the bodies are resting on each other. Can only exist for bodies with finite density.

Stability Against Impact: When the relative distance between bodies in a system are bounded away from each other for all future time.

Sundman's Inequality: An inequality that relates angular momentum, kinetic energy and potential energy.

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Biographical Sketch

Daniel Jay Scheeres (born in 1963 in Royal Oak, Michigan, USA) has a Bachelor's of Science in Letters and Engineering from Calvin College (1985), a Bachelor's and Master's of Science in Aerospace Engineering from The University of Michigan (1987 and 1988, respectively), and a PhD. in Aerospace Engineering from The University of Michigan (1992) where he studied with Nguyen Xuan Vinh. Since 2008 he has been the A. Richard Seebass Endowed Chair Professor in the Department of Aerospace Engineering Sciences at The University of Colorado Boulder. Prior to this he held academic positions at the University of Michigan and Iowa State University. Prior to that he was a Senior Member of the Technical Staff at the Jet Propulsion Laboratory / California Institute of Technology. He is past chair of the American Astronomical Society's Division on Dynamical Astronomy and the vice-president of the Celestial Mechanics Institute. His research interests include celestial mechanics of distributed bodies, the mechanics of comets and asteroids, the navigation and dynamics of spacecraft, and the long-term evolution of orbit debris.